

E-COMMERCE PLATFORM WITH **AUGMENTED REALITY**



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ABSTRACT

With rapid growth of online shopping, users increasingly struggle to visualize how products will look and fit in their real environment. Items such as rugs, furniture, home decorations and accessories are often purchased based on static images and dimensions, which can lead to confusion and product returns when the item that is delivered does not match the user expectations. Traditional E-commerce interfaces do not provide an immersive aware review of the product, which limits the customer's ability to make informed decisions.

This project proposes a mobile-based e-commerce platform enhanced with Augmented Reality (AR) to provide users with a more realistic and interactive shopping experience. This system lets users explore products, view detailed information and then virtually place 3D models of selected items in their surroundings using their smartphone's camera. The application is built using Flutter for cross-platform mobile development and is linked to a cloud backend for user identification, product and order data management and media storage. An AR framework enables AR functionality by leveraging device features such as surface detection and real-time rendering of 3D objects.

By allowing users to preview things in their own surroundings before purchasing, the system hopes to boost user confidence, improve decision making and reduce the likelihood of returns. The platform is a smart, user-friendly solution that blends e-commerce with augmented reality technology to provide more interesting and trustworthy online purchasing experiences.

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CHAPTER # 1
SOFTWARE PROJECT MANAGEMENT PLAN
(SPMP)

1.1 INTRODUCTION

E-commerce has become a common way for people to buy products, especially for categories such as furniture, rugs, home decorations and accessories. Online platforms allow the users to compare prices, browse catalogues and place orders without visiting a physical store. However, customers usually see only static images, written descriptions and a few user reviews. For the products that are heavily on size, color and placement in a room, it is still difficult to judge how an item will look and fit in the user's own environment, which can lead to uncertainty and product returns.

Figure 1.1 provides a visual representation of how AR technology improves the online shopping experience by allowing users to visualize the products within their own environment before purchase, reducing uncertainty and likelihood of returns.



Figure 1.1 : E-Commerce with Augmented Reality (AR)

Augmented Reality (AR) helps reduce this gap between the online catalogue and the real world. AR systems overlay digital content, such as 3D objects or labels on top of live camera view of a phone or tablet, allowing the users to see virtual items inside their real surroundings. This technology is already used in social media filters, navigation, games and education to make information more interactive and context aware.

When AR is combined with e-commerce, customers now can visualize products at real scale in their own rooms before buying. For example, a user can place a virtual sofa on the living room floor, move it around and see how it looks with existing furniture and wall colors. Industry reports indicate that such AR experiences can increase user engagement, support more confident purchase decisions and help reduce the likelihood of returns.

1.1.1 Existing E-Commerce Systems

1.1.1 Existing E-Commerce Systems With AR

Several well-known e-commerce applications already include AR features for furniture and home décor:

- **IKEA PLACE / IKEA APP:** IKEA's AR solution allows users to place true to scale 3D models of IKEA furniture in their rooms using a smartphone camera. It is limited to IKEA products and markets where IKEA operates.

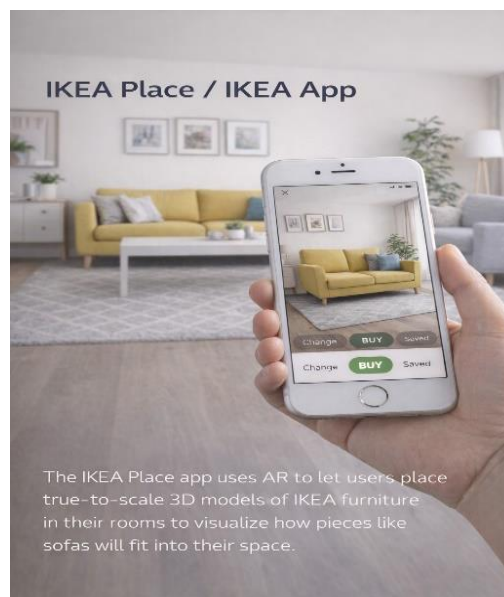


Figure 1.2 : IKEA Place / IKEA App

Figure 1.2 shows The IKEA Place app uses AR to let users place 3D models of IKEA furniture in their rooms to visualize how pieces like sofas will fit into their space.

- **AMAZON AR VIEW:** The Amazon app offers an AR mode for selected items, enabling the users to see certain items inside their own environment. It is only available for a subset of products.



Figure 1.3 : Amazon AR View

Figure 1.3 shows that it allows users to visualize selected products in their own space at true scale using AR. In this figure, a person is holding a smartphone to view a brown armchair placed within a real living room setting.

- **HOZZ:** This app includes AR tools for visualizing tiles, décor and other items on floors and walls under real lighting conditions. It supports only selected product types sold through Houzz.



Figure 1.4 : Houzz App

Figure 1.4 shows this app uses AR to let users visualize furniture and décor in their own space. In this figure, a person is holding a smartphone displaying a 3D AR model of a modern living room setup, including a navy sectional sofa and a round wooden coffee table.

1.1.2 Existing E-Commerce Systems Without AR

Traditional e-commerce platforms such as Amazon, Daraz, AliExpress and many local online stores mainly rely on 2D images, text descriptions, customer reviews and specifications. They are effective for listing large catalogues and managing orders, but they do not allow users to visualize products inside their own rooms. Because of this, customers may continue to struggle to understand the true scale and appearance of things, resulting in inaccurate size expectations, problems matching colors and a reliance on guessing and reviews rather than visual confirmation.

1.1.3 Project Overview

The main goal of this project is to develop a mobile e-commerce application with built-in AR visualization features. The app will allow the users to browse different product categories (such as furniture, carpets, home decorations and accessories), view detailed product information and then use AR to virtually place selected items in their real environment using the phone's camera.

The system will be implemented using Flutter for cross-platform mobile development. The backend will use Firebase services, including Firebase Authentication for user accounts, Cloud Firestore for storing products, user and order data and Firebase Storage for product images and 3D AR models.

The key features of the project include:

- User registration and login
- Browse the product and search it by category
- Product details with images, description, price and AR preview option
- Shopping cart and order placement
- AR based visualization of selected products in the user's environment
- Order history and profile management

1.1.4 Project Deliverables

The deliverables of my project are as follows:

- Software Project Management Plan
- Software Requirements Specification
- Software Design Description
- Software Text Documentation
- Mobile Application Interfaces
- Augmented Reality Visualization Module

- Backend Configuration
- Source Code
- Final Working Software (demonstrating AR-enabled e-commerce flow)

1.2 PROJECT ORGANIZATION

1.2.1 Software Process Model

I chose the Iterative Incremental Model for development of the E-Commerce Platform with Augmented Reality due to the following reasons:

- The system consists of multiple modules (product catalog, authentication, AR viewer) that can be developed and improved in small increments.
- AR features often require experimentation and tuning, which fits well with an iterative approach where feedback is used to refine each increment.
- The Iterative Incremental Model allows delivering a basic working e-commerce app in early iterations (without AR) and then gradually integrating and improving the AR module in later iterations.

As shown in *Figure 1.5*, the Iterative Incremental Software Process Model organizes the project into repeated cycles. Each iteration delivers a working subset of the system and later iterations add more functionality and refinement.

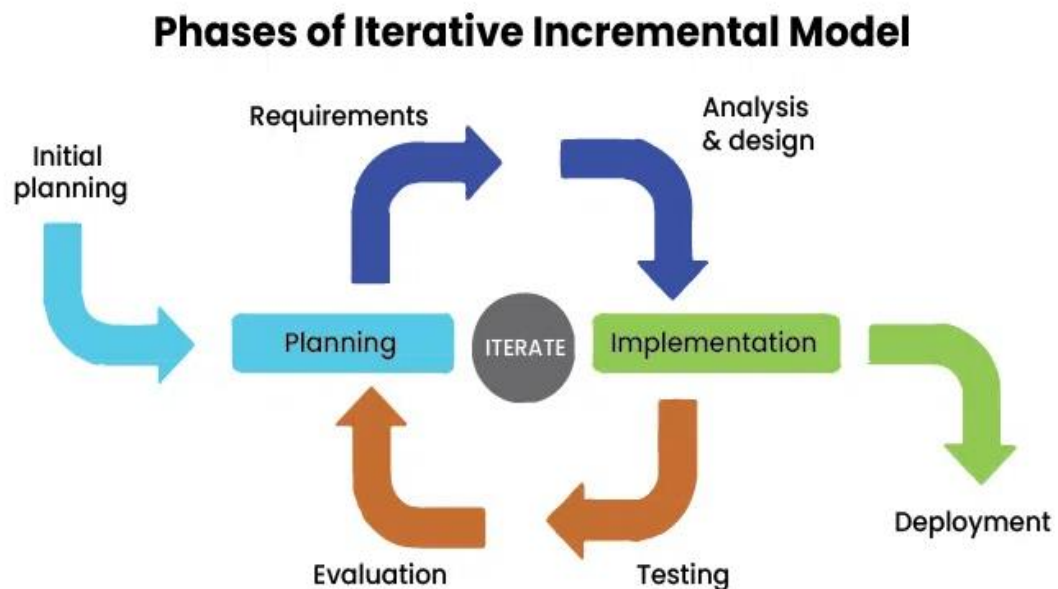


Figure 1.5 : Iterative Incremental Software Process Model

1.2.2 Roles and Responsibilities

Developer (Student):

- Select and study suitable AR tools and Flutter plugins for integrating ARCore with the mobile app.
- Design and implement the Flutter UI for product browsing, AR preview, cart and orders.
- Configure and integrate Firebase services (Authentication, Firestore, Storage) with the mobile app.
- Implement core e-commerce logic, including product management, cart operations and order handling.
- Implement the AR visualization module, ensuring that the models can be placed, moved and viewed realistically in the user's environment.
- Perform unit and integration testing of the critical features and fix bugs during development.

Supervisor:

- Provide guidance on the overall system design, especially the use of AR for e-commerce and the choice of technologies.
- Review the project documents and provide feedback timely.
- Help resolve conceptual or technical issues when the student faces major roadblocks.

1.2.3 Tools and Techniques

Tools Used:

- Flutter and Dart for developing mobile application
- Android Studio as an IDE
- Firebase Authentication, Cloud Firestore, Firebase Storage for backend services
- AR Flutter Plugin to enable augmented reality
- ProjectLibre for project scheduling, task planning and Gantt Chart generation
- Visual Paradigm / draw.io for drawing UML diagrams
- Git and GitHub for version control and backups
- MS Word with IEEE-style format for preparing project documentation

Techniques Used:

- Iterative and Incremental Development
- Modular Design
- User-Centered UI Design

- AR Plane Detection and Object Placement
- State Management in Flutter
- Basic Security Practices

1.3 PROJECT MANAGEMENT PLAN

1.3.1 Tasks

The project consists of the following main phases:

- Requirements Analysis Phase
- System Design Phase
- Prototype and UI Design Phase
- Development Phase
- Testing & Validation Phase

1.3.2 Description

The first step is the Requirements Analysis which helps identify key functional and non-functional requirements of the AR enabled e-commerce mobile application, such as product browsing, AR visualization, cart management and performance constraints on mobile devices.

The next phase is System Design Phase where the overall architecture is defined, the interaction between the mobile app and backend is planned, data models are designed and the AR module's integration points are specified.

The Prototype and UI Design then focus on creating and refining the main app screens in Flutter, designing user flows for browsing products, viewing details, launching AR view and placing orders so that the interface is intuitive and user-friendly.

In the Development Phase, the planned features are implemented in iterations, including user authentication, product catalog, AR product placement using device camera, cart operations and order handling.

Testing and Validation phase is concerned with checking the correctness and reliability of all the modules through functional testing, AR behavior testing on real devices, usability feedback from users and fixing issues to ensure that the system meets the defined requirements before the deployment.

1.3.3 Project Plan

Figure 1.6 presents the project plan in the form of Gantt chart. The chart groups tasks into requirement analysis, system design, prototype and UI design, development, testing and validation activities and shows how these tasks are scheduled over the documentation phase of the project.

		Name	Duration	Start	Finish
1		Planning	10 days	10/1/25 8:00 AM	10/14/25 5:00 PM
2		Define Project Deliverables for AR E-Commerce App	2 days	10/1/25 8:00 AM	10/2/25 5:00 PM
3		Choose & Define Software Process Model	2 days	10/3/25 8:00 AM	10/6/25 5:00 PM
4		Define Roles & Responsibilities	2 days	10/7/25 8:00 AM	10/8/25 5:00 PM
5		Define Tools & Techniques (Flutter, Firebase, AR plugin, etc.)	2 days	10/9/25 8:00 AM	10/10/25 5:00 PM
6		Define Project Timeline (docs + implementation)	1 day	10/13/25 8:00 AM	10/13/25 5:00 PM
7		Verify SPMP	1 day	10/14/25 8:00 AM	10/14/25 5:00 PM
8		Analysis	15 days	11/3/25 8:00 AM	11/21/25 5:00 PM
9		Describe Scope & Objectives of AR E-Commerce System	2 days	11/3/25 8:00 AM	11/4/25 5:00 PM
10		Define Functional & Non-Functional Requirements	4 days	11/5/25 8:00 AM	11/10/25 5:00 PM
11		Define Use Cases (customer, admin, AR preview, orders, etc.)	3 days	11/11/25 8:00 AM	11/13/25 5:00 PM
12		Draw Use Case Diagram	2 days	11/14/25 8:00 AM	11/17/25 5:00 PM
13		Draw System Sequence Diagrams	2 days	11/18/25 8:00 AM	11/19/25 5:00 PM
14		Draw Domain Model	1 day	11/20/25 8:00 AM	11/20/25 5:00 PM
15		Verify SRS	1 day	11/21/25 8:00 AM	11/21/25 5:00 PM
16		Software Design	20 days	12/1/25 8:00 AM	12/26/25 5:00 PM
17		Create Requirements Traceability Matrix	2 days	12/1/25 8:00 AM	12/2/25 5:00 PM
18		Design System Architecture (Flutter app + Firebase + AR module)	3 days	12/3/25 8:00 AM	12/5/25 5:00 PM
19		Describe System Interfaces (screens, API, Firebase collections, AR)	3 days	12/8/25 8:00 AM	12/10/25 5:00 PM
20		Describe System Components (UI, business logic, AR layer, backend)	3 days	12/11/25 8:00 AM	12/15/25 5:00 PM
21		Draw Sequence Diagrams (key flows: browse AR cart order)	3 days	12/16/25 8:00 AM	12/18/25 5:00 PM
22		Draw Class Diagram	2 days	12/19/25 8:00 AM	12/22/25 5:00 PM
23		Design User Interfaces (Flutter screen layouts for AR e-commerce)	3 days	12/23/25 8:00 AM	12/25/25 5:00 PM
24		Verify SDD	1 day	12/26/25 8:00 AM	12/26/25 5:00 PM
25		Testing and Validation	15 days	1/15/26 8:00 AM	2/4/26 5:00 PM
26		Define Test Approach Used (black-box, device testing, AR checks)	2 days	1/15/26 8:00 AM	1/16/26 5:00 PM
27		Make Test Plan (features to test, environment, devices, schedule)	3 days	1/19/26 8:00 AM	1/21/26 5:00 PM
28		Write Test Cases (login, catalog, AR placement, cart, orders, etc.)	7 days	1/22/26 8:00 AM	1/30/26 5:00 PM
29		Verify STD	3 days	2/2/26 8:00 AM	2/4/26 5:00 PM

Figure 1.6 : Project Management Plan

1.3.4 Project Timeline

Figure 1.7 summarizes the overall timeline of the project from the start of documentation work in October 2025 to the completion of implementation and testing in June 2026. It highlights the separation between documentation-focused period and later implementation-focused period.

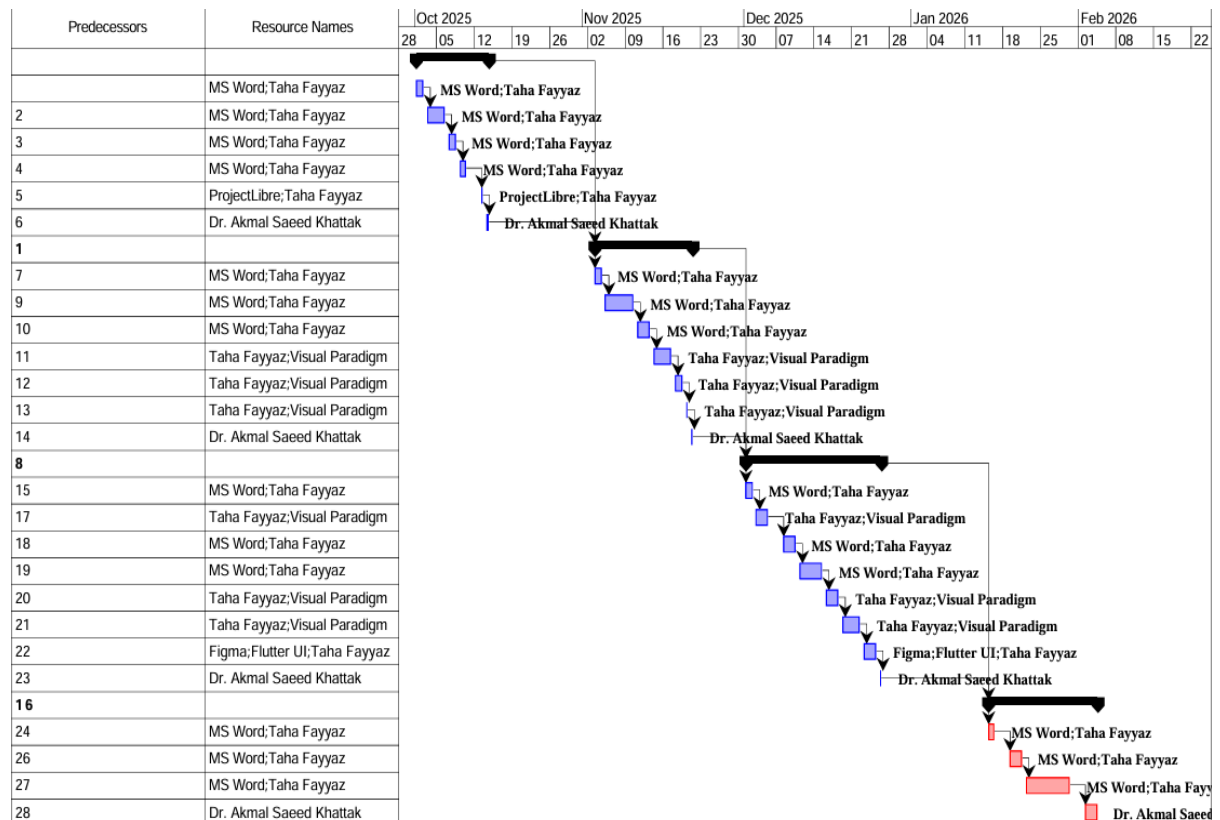


Figure 1.7 : Project Management Timeline

CHAPTER # 2
SOFTWARE REQUIREMENT SPECIFICATIONS
(SRS)

2.1 INTRODUCTION

This chapter defines the software requirements for E-Commerce Platform with Augmented Reality. It clarifies the problem being solved, why the system is needed, what the project intends to deliver, and the boundaries under which the system will operate. The chapter also states key constraints and assumptions and presents the overall workflow of the proposed system. This chapter also presents the system use cases and their descriptions for clarity.

2.1.1 Problem Statement

Conventional E-commerce applications mostly rely on static images of the products, basic dimensions and text descriptions. For items such as furniture, rugs and home decorations, this makes it difficult for the users to estimate true placement, scale and visual fit in their own environment. As a result, customers make uncertain selections, which leads to dissatisfaction and increased return rates.

The challenge is to create an e-commerce system that uses Augmented Reality to allow users to preview products in their real-world surroundings before making a purchase while simultaneously supporting regular shopping functions.

2.1.2 Motivation

The motivation for this project is to simplify internet shopping for environmentally friendly products. AR-based visualization can boost user experience and confidence by providing a realistic glimpse at near-actual scale and positioning. An informed decision-making process is anticipated to:

- Increase user trust in product selection.
- Improve shopping engagement.
- Reduce avoidable returns caused by size/fit mismatch.
- Deliver a modern and competitive shopping experience on mobile devices.

2.1.3 Objectives

The project's key objectives include:

- Creating a mobile-based e-commerce application for product browsing and purchasing.
- Provide an AR preview feature to place products 3D model into user's real environment.
- Support core shopping flows: authentication, product browsing/search, product details, cart, checkout and order placement.

- Store and manage product, user and order data using cloud backend (Firebase services).
- Ensure a user-friendly interface and consistent performance on compatible mobile devices.

2.1.4 Scope

The project's scope comprises the development of a cross-platform mobile application covering the following areas:

- User registration, login and profile management.
- Product catalog browsing and category-based navigation.
- Product detail view (images, description, price and AR preview option).
- AR-based visualization for products.
- Cart management and checkout flow.
- Order placement and order history view.
- Cloud data storage for product images, product metadata and 3D assets.

2.1.5 Constraints

The system is subject to the following limitations:

- **Device Compatibility:** AR preview requires devices that support ARCore/ARKit. Devices lacking AR capabilities can use the app but not the AR preview.
- **Performance:** Real-time AR rendering depends on device GPU; low-end devices may show reduced performance.
- **Network Dependence:** Internet connectivity is required to fetch product catalogs, images and 3D models from cloud storage.
- **3D Asset Availability:** AR preview is only possible for products that have valid, optimized 3D models uploaded to storage.
- **Lighting/Surface Limitations:** AR plane detection depends on environment lighting and visible surfaces; inaccurate detection may affect preview quality.
- **Project Timeline:** Development and documentation must follow the semester/FYP schedule, limiting scope expansion.

2.1.6 Assumptions

The project is designed with the following assumptions:

- Users have a smartphone with camera and basic mobile skills.
- Users have stable internet for surfing and checkout.
- Product catalog data (price, description, images, and stock availability) is maintained correctly in the backend.
- 3D models are provided in supported formats and are optimized for mobile rendering.
- Users will use AR preview in suitable conditions (adequate lighting and visible floor/table surface).
- The system aims to integrate with a third-party payment gateway for online payments. If gateway integration is not completed during the prototype phase, the system will enable Cash-on-Delivery (COD) or simulated payment confirmation.

2.1.7 System Workflow Diagram

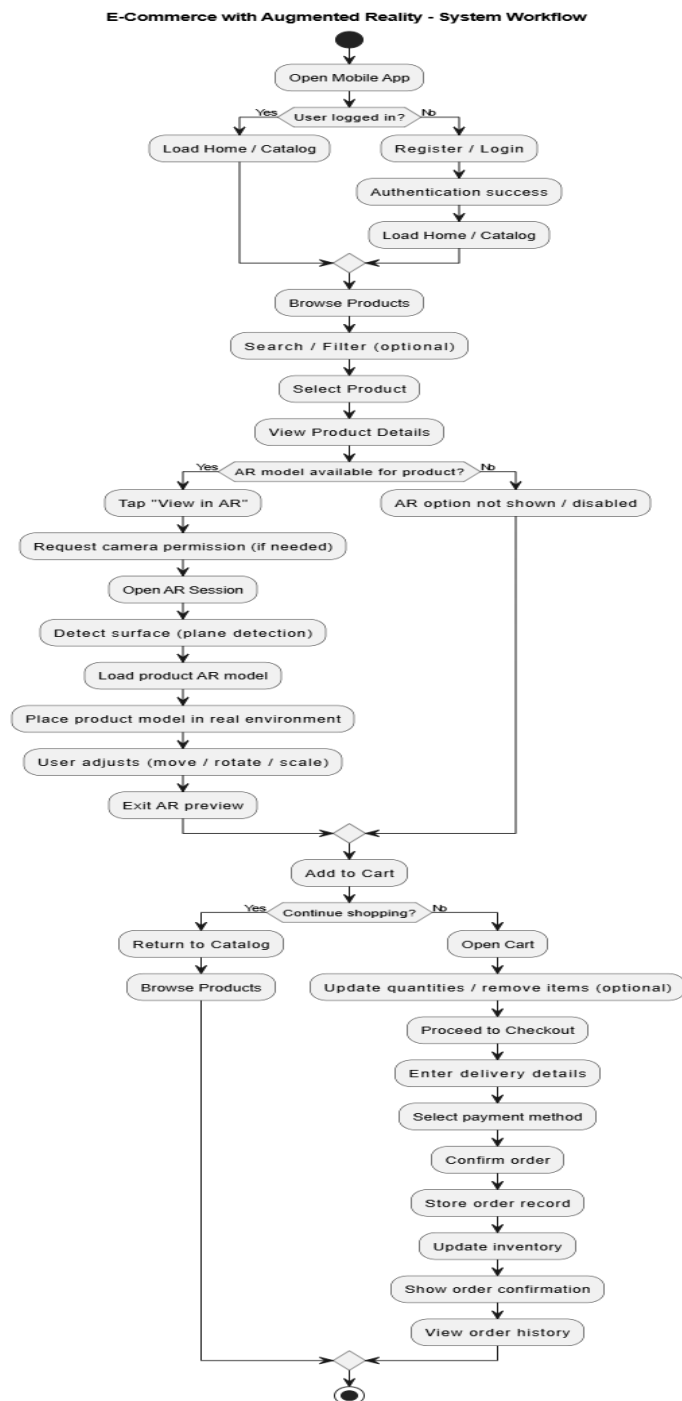


Figure 2.1 : System Workflow Diagram

Figure 2.1 depicts how the system works from the beginning to the end of an order. The user launches the app, logs in (if necessary), and browses the product catalog using the search and filter features. When a 3D model of the product is accessible, the user can preview it after examining its details. Otherwise, the AR option is disabled. The user then adds the goods to the

cart, goes to the checkout, enters delivery information, chooses a payment option and confirms the order. Finally, the order is stored in the backend (Firebase), where the user may view the confirmation and order history.

2.2 OVERALL DESCRIPTION

2.2.1 Product Perspective

The E-Commerce Platform with Augmented Reality is a mobile-based standalone system that combines traditional e-commerce functions with an AR visualization module. Customers can explore products, examine extensive information and preview products in their real environment utilizing augmented reality, as well as place purchases via a checkout procedure. A dedicated admin interface is also provided for managing product data, AR assets and order records. The solution uses cloud services for authentication, data storage and media hosting.

Relationship with external components/systems:

- **Firebase Authentication (External Service):** Enables user registration, login and session management.
- **Cloud Firestore (External Database Service):** Stores structured data including user profiles, product catalog details, cart/order records and order status.
- **Firebase Storage (External Storage Service):** Stores product images and 3D models required for AR-enabled products.
- **AR Framework (ARCore/ARKit via AR plugin):** Handles AR session creation, plane detection, tracking and rendering of 3D product models in the real environment.
- **Device Hardware (Camera and Sensors):** The mobile camera and sensors are required to support AR preview and environment tracking.
- **Payment Gateway (External Service):** Used to authorize/capture online payments. COD/simulated confirmation may be used as a prototype fallback.

Internal module included in the product:

- **Admin Interface Module:** A specialized interface for administrators to manage items, upload photos and 3D models and monitor/update orders. It connects to the same cloud backend (Firestore/Storage) as the customer application.

2.2.2 User Characteristics

1. Customer / Buyer (Primary User):

- Can register/login, browse products, search/filter, view product details, and place orders.

- Uses AR preview for products to visualize product placement and approximate size before purchase.
- A straightforward and guided checkout flow with a few steps is necessary.

2. Administrator (Admin Interface User):

- Use the admin interface to manage system data and operations.
- Can add/update/delete products and can upload product images, and can associate supported 3D models with AR-enabled products.
- Can view orders and update status (pending/confirmed/shipped/delivered) based on the implemented workflow.

2.3 SPECIFIC REQUIREMENTS

2.3.1 External Interface Requirements

2.3.1.1 User Interfaces

- **Customer Mobile Application:**
 - The system must offer Login/Signup interfaces with validation for needed fields.
 - The system must provide a Catalog interface that includes product name, price and thumbnail.
 - The system shall provide Search by product name and Filter by category.
 - The system shall provide Product Detail interface showing images, name, price, description and an AR Preview only when a 3D model is linked.
 - The system should have a Cart interface that displays items, quantities, subtotal and total.
 - The system should provide a checkout interface to collect recipient information, including name, delivery address and payment method.
 - The system must include order confirmation and order history interfaces that display order ID, summary and status.
- **Admin Interface:**
 - The system shall provide Admin Login.
 - The system shall provide Product Management (add/update/delete).
 - The system shall provide Media Management to upload product images and upload/associate 3D models.
 - The system should offer Order Management to see and adjust order status.

2.3.1.2 Hardware Interfaces

- **Customer side:** Android/IOS smartphone with camera.
- **AR Preview:** Device shall be ARCore/ARKit supported.
- **Admin Side:** Laptop/Desktop (or Mobile) capable of running the admin interface.

2.3.1.3 Software Interfaces

- **Mobile App:** Flutter (Dart).
- **Backend:** Firebase Authentication, Cloud Firestore, Firebase Storage.
- **AR:** ARCore/ARKit via Flutter AR plugin.

2.3.1.4 Communication Protocols

- All client-backend communication must take place over HTTPS.
- The system must use security controls and authenticated access to access images/3D models stored on Firebase Storage.

2.3.2 Functional Requirements

FR-01 User Authentication

- The system shall allow users to register and login using Authentication service.
- The system shall prevent access to cart/checkout/order history when the user is not authenticated.

FR-02 Product Catalog

- The system shall retrieve products from the product catalog and display them in the catalog.
- Each product shall include at minimum: name, price, category, description, stockQuantity, images.

FR-03 Search and Filter

- The system shall support keyword search on product name.
- The system shall support filtering by category.

FR-04 Product Details

- The system shall display full product details including images, price and description.
- The system shall display the AR Preview option only if the product has valid 3D model reference.

- The system shall display product availability as In Stock / Out of Stock based on stockQuantity.

FR-05 AR Preview

- When AR Preview is selected, the system shall request camera permission if not granted.
- If permission is denied or device is unsupported, the system shall disable AR preview and display a reason.
- The system shall initialize the AR session, load 3D model and allow the user to place and adjust it in the real time environment.

FR-06 Cart Management

- The system shall allow users to add items to cart, update quantities, and remove items.
- The system shall compute subtotal and total based on cart contents.
- The system shall prevent users from adding quantity greater than the available stockQuantity.

FR-07 Checkout and Order Placement

- The system shall validate delivery fields (name, phone, address) before confirming an order.
- The system shall create and store an order record upon confirmation.
- The system shall verify stock availability before confirming an order.
- On successful order placement, the system shall decrease stockQuantity for each ordered product.

FR-08 Order History

- The system shall allow users to view only their own orders and current order status.

FR-09 Customer Profile Management

- The system shall allow users to view and update their profile information.

FR-10 Admin Product Management

- The system shall allow an administrator to add/update/delete products.
- The system shall allow admin to update stockQuantity for products.

FR-11 Admin Media Management

- The system shall allow the administrator to upload product images and associate 3D models with AR-enabled products.

FR-12 Admin Order Management

- The system shall allow admin to view all orders and update order status.

2.3.3 Non-Functional Requirements

Usability:

- The system shall allow a user to reach Product Details from Catalog within ≤ 3 taps.
- The system shall allow a user to complete (Add to Cart to Checkout to Confirm Order) within ≤ 6 taps (excluding typing).

Performance:

- The system shall load the initial Catalog screen within ≤ 5 seconds on a standard 4G/Wi-Fi connection.
- The system should open Product Details within ≤ 4 seconds if data is cached, and ≤ 6 seconds otherwise.

Reliability:

- If a network request fails, the system shall display an error message within ≤ 10 seconds and offer a retry option.

Security:

- The system shall enforce HTTPS for all communication.
- Firestore rules shall ensure: users can read/write only their own profile/orders.
- Admin-only operations (product/order updates) shall be restricted to authorized admin accounts.

Compatibility:

- The system should run on Android and iOS devices that support the Flutter SDK.
- Non-AR devices can access all e-commerce capabilities, except AR preview.

Maintainability:

- The system should divide modules (auth, catalog, AR, cart, orders, admin) so that changes in one do not necessitate redesign of other modules.

2.4 SOFTWARE PRODUCT FEATURES

The primary software product features of this system include:

- **User Account Management**
Cloud-based authentication enables registration, login and basic profile management.
- **Product Catalog Browsing**
Product listing with category-wise navigation.
- **Search and Filtering**

Use Keyword search and category filters to easily find goods.

- **Product Detail View**

The product interface includes photographs, descriptions, prices, stock status and an AR availability indicator.

- **Augmented Reality (AR) Product Preview**

AR product visualization enables users to insert 3D models in real-world environments.

- **Shopping Cart Management**

Features include add-to-cart, quantity changes, item removal and auto-calculation of subtotal and total.

- **Checkout and Order Placement**

Enter delivery details, select payment options, confirm orders and create records in the backend.

- **Order History and Tracking**

View past orders and their current status.

- **Admin Interface for Product and Order Management**

Admins can manage items, pictures, AR models and order statuses.

2.5 SOFTWARE SYSTEM ATTRIBUTES

2.5.1 Reliability

- The system must maintain product, cart and order data integrity during regular operations.
- The system must ensure that confirmed orders are not lost in Firestore.
- The system should handle AR initialization or tracking failure gracefully, returning the user to product interface with an explanation.

2.5.2 Availability

- Customers can access the system 24 hours a day, 7 days a week, depending on Firebase service availability and internet connectivity.
- The admin interface shall be available during typical business hours but can also be hosted for 24/7 access if needed for deployment.

2.5.3 Security

- The system shall use HTTPS for all client–backend communication.
- The system shall enforce Firebase security standards, allowing users to view just their own profile and orders. Product and order management requires admin credentials.

2.5.4 Maintainability

- The system shall follow a modular structure (Authentication, Catalog, AR Preview, Cart, Orders, Admin).
- The system shall allow replacement or upgrade of AR plugins and Firebase configurations without requiring a full redesign of the application.

2.5.5 Portability

- The customer application should be compatible with Android and iOS platforms using the Flutter SDK.

2.5.6 Performance

- The system should load the initial product catalog within 5 seconds using a typical 4G/Wi-Fi connection.

2.6 DATABASE REQUIREMENTS

The system relies on Firebase Cloud Firestore for structured data and Firebase Storage for media assets, including product photos and AR 3D models. The database stores application data consistently and securely for clients and administrators.

Firestore will store the following:

- **User Records**
User ID, name, email, phone number (optional), and basic profile information.
- **Product Catalog**
Product ID, title/name, category, price, description, image URLs, stockQuantity/availability status and AR model reference (only for AR-supported products).
- **Cart Data**
Cart data shall be stored per user in Firestore to support session continuity.
- **Order Records**

Order ID, user ID, ordered items, quantity, price totals, delivery details, timestamp, and order status (e.g., pending/confirmed/shipped/delivered).

Firebase Storage stores:

- **Product Images**
(JPEG/PNG) files connected to URLs in Firestore.
- **3D AR Models**
linked through model references stored in Firestore.

Data retention and consistency:

- Orders are stored permanently for record and tracking purposes.
- Storage access shall be controlled through Firebase security rules so that customers cannot modify product catalog data, while admin can manage products and assets.

2.7 USE CASES

2.7.1 Use Cases List

Customer

Use Case ID	Use Case Name
UC-01	Create Customer Account
UC-02	Explore Product Catalog (Browse/Search/Filter)
UC-03	Review Product Details
UC-04	Visualize Product in Augmented Reality
UC-05	Maintain Shopping Cart
UC-06	Checkout and Submit Order
UC-07	Review Order History
UC-08	Maintain Customer Profile

Table 2.1: Customer Use Cases

Admin

Use Case ID	Use Case Name
UC-09	Maintain Product Catalog
UC-10	Maintain Product Media Assets (Images & 3D Models)
UC-11	Maintain Inventory/Stock
UC-12	Process Orders (View/Update Status)

Table 2.2: Admin Use Cases

2.7.2 Use Case Diagram

Figure 2.2 depicts the system's major interactions with its two main actors: the customer and the administrator. The customer completes core purchasing actions such as creating an account, browsing the product catalog, examining product descriptions, previewing goods in augmented reality, placing orders through checkout, viewing order history and updating profile information. The administrator handles backend activities such as product catalog management, uploading product photos and 3D models, updating inventory/stock, and processing customer orders by changing order status.

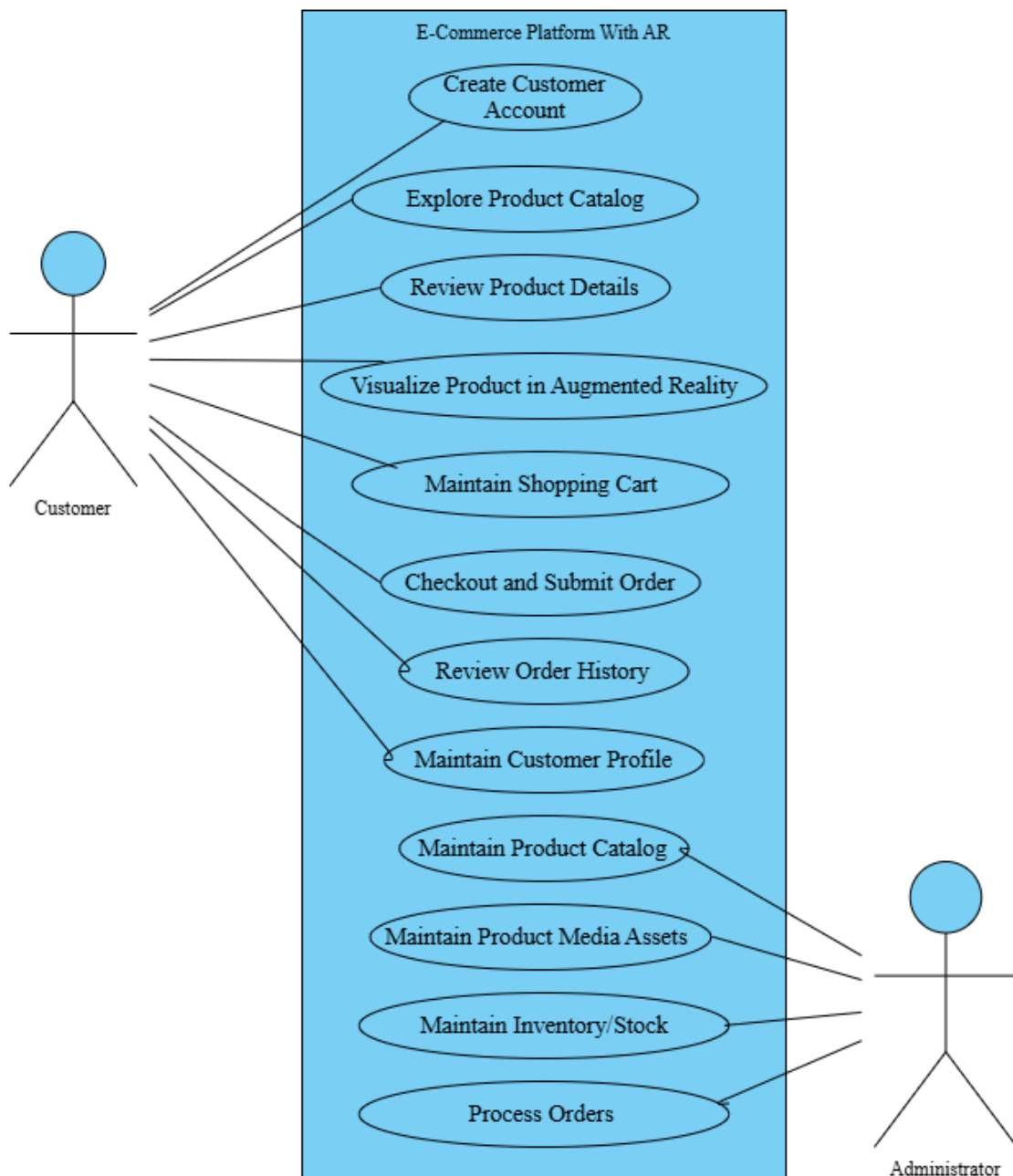


Figure 2.2: Use Case Diagram

2.7.3 Use Cases Description

UC-01 Create Customer Account

Table 2.3 explains how customers can create an account by completing registration information. The system validates information, establishes an account, sets up the customer profile, and provides access to the customer module.

Use Case ID	UC-01
Use Case Name	Create Customer Account
Primary Actor	Customer
Stakeholders & Interests	Customer: create account quickly and securely. System Owner: maintain valid user identities for ordering and tracking.
Preconditions	1. App is accessible. 2. Internet connectivity is available.
Postconditions	1. User account created. 2. Customer profile is initialized and linked to the account. 3. Customer session becomes active.
Main Success Scenario	1. Customer selects Sign Up. 2. System presents a registration form. 3. Customer inputs essential information (name, email, password). 4. Customer submits the form. 5. System validates data and creates the account. 6. System initializes the customer profile. 7. System redirects customer to Catalog/Home.
Extensions/Alternative Flows	3A Invalid data: System displays validation message; user retries. 5A Email already exists: System prompts login/reset. 5B Network/auth failure: System displays error and permits retry.
Frequency of Occurrence	Low (new customers).
Special Requirements	Password must be masked; error messages must be clear.
Technology	Flutter UI; Firebase Authentication; Firestore (Users).
Inputs	Name, email, password; submit action.
Outputs	1. Account created. 2. profile saved. 3. access to customer module.

Table 2.3: UC-1 Create Customer Account

UC-02 Explore Product Catalog

Table 2.4 describes how customers browse the catalog and use search and category filters to discover products. The system retrieves matching products from the product catalog and displays them in the catalog view.

Use Case ID	UC-02
Use Case Name	Explore Product Catalog
Primary Actor	Customer
Stakeholders & Interests	Customer: find products quickly. System Owner: products must appear with correct price and stock status.
Preconditions	1. App is open. 2. Internet connectivity is available.
Postconditions	Matching product list is displayed based on browse/search/filter selection.
Main Success Scenario	1. Customer opens Catalog screen. 2. System fetches products. 3. Customer browses categories. 4. Customer enters a keyword and/or selects category filter. 5. System updates list based on criteria. 6. System displays the filtered/browsed product list.
Extensions/Alternative Flows	2A Load failure: System shows error and retry. 5A No matches: System shows (No products found).
Frequency of Occurrence	High.
Special Requirements	Catalog should load within defined performance limits.
Technology	Firestore queries; cached image URLs (Storage).
Inputs	Search keyword; selected category; browse actions.
Outputs	Updated catalog list.

Table 2.4: UC-2 Explore Product Catalog (Browse/Search/Filter)

UC-03 Review Product Details

Table 2.5 describes how a customer opens a product to view its details, including price, images, description, stock status, and AR availability.

Use Case ID	UC-03
Use Case Name	Review Product Details
Primary Actor	Customer
Stakeholders & Interests	Customer: understand product before purchase. System Owner: show correct stock and AR availability.
Preconditions	1. Product exists in catalog. 2. Customer selects a product.
Postconditions	1. Product details are displayed. 2. Stock and AR indicators are shown.
	1. Customer selects a product from catalog. 2. System retrieves product details.

Main Success Scenario	3. System displays images, description, price. 4. System displays stock status (In Stock/Out of Stock). 5. System enables AR option only if model reference exists and device supports AR.
Extensions/Alternative Flows	2A Product removed/not found: System shows message and returns to catalog. 3A Media unavailable: System shows placeholder and continues.
Frequency of Occurrence	High.
Special Requirements	AR button must be disabled/hidden when unavailable or unsupported.
Technology	Firestore (Product); Storage URLs (images/model ref).
Inputs	Product selection.
Outputs	Product detail interface displayed.

Table 2.5: UC-3 Review Product Details

UC-04 Visualize Product in Augmented Reality

Table 2.6 describes how a customer previews a product using AR. The system starts an AR session, loads the 3D model, and renders it on a detected surface if the device supports AR and permissions are granted.

Use Case ID	UC-04
Use Case Name	Visualize Product in Augmented Reality
Primary Actor	Customer
Stakeholders & Interests	Customer: preview product at approximate scale/placement. System Owner: stable AR experience with clear failure handling.
Preconditions	1. Product has valid 3D model reference. 2. Device supports ARCore/ARKit. 3. Camera permission granted (or requestable).
Postconditions	AR preview shown OR user receives clear failure message and returns to product details.
Main Success Scenario	1. Customer taps View in AR. 2. System requests camera permission (if needed). 3. System initializes AR session. 4. System downloads/loads 3D model (cache if available). 5. System detects surface and displays placement guide. 6. Customer places and views model in environment. 7. Customer exits AR and returns to product details.
	2A Permission denied: System shows message; AR not started.

Extensions/Alternative Flows	3A Unsupported device: System shows (AR not supported). 4A Model load fails: System displays an error message and returns to details. 5A Surface not detected: System provide guidance and offer the option to retry or exit.
Frequency of Occurrence	Medium.
Special Requirements	AR failure must not crash; guidance must be visible if tracking is weak.
Technology	ARCore/ARKit via Flutter AR plugin; Firebase Storage model download.
Inputs	(View in AR) action; permission response.
Outputs	AR preview session OR explanatory error message.

Table 2.6: UC-4 Visualize Product in Augmented Reality

UC-05 Maintain Shopping Cart

Table 2.7 shows how customers can add products to their shopping, adjust amounts, or remove items. The system recalculates totals and enforces stock quantity limits.

Use Case ID	UC-05
Use Case Name	Maintain Shopping Cart
Primary Actor	Customer
Stakeholders & Interests	Customer: manage intended purchases. System Owner: enforce stock limits and maintain correct totals.
Preconditions	1. Customer is logged in. 2. Product exists in catalog.
Postconditions	1. Cart is updated and saved for the customer. 2. Subtotal/total updated.
Main Success Scenario	1. Customer selects a product and presses the Add to Cart button. 2. System checks stock Quantity. 3. System adds item (or increments quantity). 4. Customer opens Cart screen. 5. Customer changes quantity or remove items. 6. System recalculates totals and saves updated cart.
Extensions/Alternative Flows	2A Out of stock: System blocks add and shows message. 5A Quantity exceeds stock: System blocks update and shows limit. 6A Network failure: System shows error and retry; cart update not confirmed.
Frequency of Occurrence	High.
Special Requirements	Quantity updates must respect stockQuantity and never exceed it.

Technology	Firestore (Cart per user); Firestore (Product stockQuantity).
Inputs	Add/remove actions; quantity changes.
Outputs	Updated cart items and totals.

Table 2.7: UC-5 Maintain Shopping Cart

UC-06 Checkout and Submit Order

Table 2.8 shows how customers complete checkout by entering delivery information and confirming their order. The system validates fields, rechecks stock, produces orders, and updates stock quantities.

Use Case ID	UC-06
Use Case Name	Checkout and Submit Order
Primary Actor	Customer
Stakeholders & Interests	Customer: place order successfully. System Owner: consistent order record and correct stock update.
Preconditions	1. Customer is logged in. 2. Cart is not empty. 3. Internet connectivity available.
Postconditions	1. Order is created with a unique order ID and recorded persistently. 2. stockQuantity reduced for ordered products. 3. Cart cleared.
Main Success Scenario	1. Customer opens cart and selects Checkout. 2. System presents a checkout form. 3. Customer inputs delivery information and chooses payment option. 4. Customer verifies the order. 5. System re-validates stock for all items. 6. System creates order and updates inventory. 7. System displays order confirmation with order ID.
Extensions/Alternative Flows	3A Invalid fields: System shows validation errors and prevents submission. 3B Online payment selected: System conducts the transaction with the payment gateway and receives a success or failure answer. 3B1 Payment failed/declined: System displays failure message and order is not made. 3B2 Gateway unavailable: System displays an error message and offers a retry option (or COD if supported). 5A Stock insufficient: System stops and asks customer to update cart. 6A Write/update failure: System shows error and does not confirm order.
Frequency of Occurrence	Medium.

Special Requirements	Order creation and stock update must be consistent (no partial confirmation).
Technology	Firestore (Orders); atomic update strategy at implementation level.
Inputs	Delivery details; payment option; confirm action.
Outputs	1. Order confirmation. 2. Stored order. 3. Updated stock.

Table 2.8: UC-6 Checkout and Submit Order

UC-07 Review Order History

Table 2.9 describes how a customer views previous orders and their current status. The system retrieves only the customer's orders and displays them.

Use Case ID	UC-07
Use Case Name	Review Order History
Primary Actor	Customer
Stakeholders & Interests	Customer: view previous purchases and status. System Owner: enforce access control per user.
Preconditions	Customer is logged in.
Postconditions	Customer order list (and selected order details) displayed.
Main Success Scenario	1. Customer opens Order History. 2. System retrieves the customers orders. 3. System displays list with status. 4. Customer selects an order to view details.
Extensions/Alternative Flows	2A No orders: System shows empty state. 2B Network failure: System shows error and retry.
Frequency of Occurrence	Medium.
Special Requirements	Customer must not view other users orders (security rules).
Technology	Firestore query by userId.
Inputs	Order history request; order selection.
Outputs	Order list and order details displayed.

Table 2.9: UC-7 Review Order History

UC-08 Maintain Customer Profile

Table 2.10 describes how a customer views or updates profile information used for contact and delivery purposes. The system validates and saves the updated profile information.

Use Case ID	UC-08
Use Case Name	Maintain Customer Profile
Primary Actor	Customer

Stakeholders & Interests	Customer: keep contact info accurate. System Owner: maintain valid profile data for orders.
Preconditions	Customer is logged in.
Postconditions	Profile information is updated.
Main Success Scenario	1. Customer opens Profile screen. 2. System loads profile data. 3. Customer edits fields (e.g., name/phone). 4. Customer selects Save. 5. System validates and saves the profile updates.
Extensions/Alternative Flows	4A Invalid data: System shows validation error; update not saved. 5A Network failure: System shows error and retry.
Frequency of Occurrence	Low to Medium.
Special Requirements	Required fields must not be empty; validation must be clear.
Technology	Firestore (User Profile).
Inputs	Edited profile fields; save action.
Outputs	Updated profile confirmation.

Table 2.10: UC-8 Maintain Customer Profile

UC-09 Maintain Product Catalog

Table 2.11 describes how an administrator adds, edits, or deletes product records. The system validates fields and saves the changes.

Use Case ID	UC-09
Use Case Name	Maintain Product Catalog
Primary Actor	Administrator
Stakeholders & Interests	Admin: manage products efficiently. Customers: see accurate products and pricing.
Preconditions	1. Admin is logged in via admin interface. 2. Admin has required privileges.
Postconditions	Product record created/updated/deleted.
Main Success Scenario	1. Admin opens Product Management. 2. Admin selects add/edit/delete. 3. Admin enters/updates product fields. 4. Admin submits changes. 5. System validates and saves the product changes. 6. System confirms success.
Extensions/Alternative Flows	3A Missing required data: System blocks and prompts correction. 5A Database failure: System shows error and allows retry.
Frequency of Occurrence	Medium.
Special Requirements	Only authorized admin can modify product catalog.
Technology	Admin interface; Firestore (Products).
Inputs	Product fields; add/edit/delete action.

Outputs	1. Updated product catalog. 2. Confirmation/error.
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Table 2.11: UC-9 Maintain Product Catalog

UC-10 Maintain Product Media Assets

Table 2.12 describes how an administrator uploads product images and AR 3D models. The system uploads media assets and associates them with the selected product.

Use Case ID	UC-10
Use Case Name	Maintain Product Media Assets
Primary Actor	Administrator
Stakeholders & Interests	Admin: upload correct media for products. Customers: view correct images and AR availability.
Preconditions	1. Admin is logged in. 2. Product exists. 3. Internet connectivity available.
Postconditions	1. Media uploaded to Storage. 2. Product updated with image URLs and/or model reference.
Main Success Scenario	1. Admin selects a product. 2. Admin chooses image/model upload. 3. System uploads file(s) to Storage. 4. System associates the uploaded media with the selected product. 5. System confirms success.
Extensions/Alternative Flows	3A Upload fails: System shows error and retry. 3B Unsupported type/size: System blocks upload and shows constraints.
Frequency of Occurrence	Medium.
Special Requirements	Model reference must be valid before enabling AR for that product.
Technology	Firebase Storage; Firestore product update.
Inputs	Selected media files; upload action.
Outputs	1. Media stored. 2. Updated product media references. 3. Confirmation/error.

Table 2.12: UC-10 Maintain Product Media Assets

UC-11 Maintain Inventory/Stock

Table 2.13 describes how an administrator updates product stockQuantity. The system validates non-negative stock values and saves the update.

Use Case ID	UC-11
Use Case Name	Maintain Inventory/Stock
Primary Actor	Administrator

Stakeholders & Interests	Admin: keep stock accurate. Customers: avoid ordering out-of-stock items.
Preconditions	1. Admin is logged in. 2. Product exists.
Postconditions	stockQuantity and/or availability status updated.
Main Success Scenario	1. Admin opens inventory/stock view. 2. Admin selects product. 3. Admin enters new stockQuantity. 4. Admin saves. 5. System validates the stock value and saves the update.
Extensions/Alternative Flows	3A Invalid value: System blocks update and shows message. 5A Database failure: System shows error and retry.
Frequency of Occurrence	Medium.
Special Requirements	stockQuantity must never be negative; admin-only access enforced.
Technology	Firestore (Products stockQuantity).
Inputs	Product selection, stockQuantity value, save action.
Outputs	1. Updated stock. 2. Confirmation/error.

Table 2.13: UC-11 Maintain Inventory/Stock

UC-12 Process Orders

Table 2.14 describes how an administrator reviews customer orders and updates order status. The system saves the updated status and confirms the change.

Use Case ID	UC-12
Use Case Name	Process Orders
Primary Actor	Administrator
Stakeholders & Interests	Admin: process fulfillment steps. Customers: receive correct order status.
Preconditions	1. Admin is logged in. 2. Orders exist (or system supports empty state).
Postconditions	Selected order status updated.
Main Success Scenario	1. Admin opens order management. 2. System lists orders with current status. 3. Admin selects an order. 4. Admin chooses new status. 5. System saves the updated order status. 6. System confirms update.
Extensions/Alternative Flows	2A No orders: System shows empty state. 5A Update fails: System shows error; status remains unchanged.
Frequency of Occurrence	Medium to High.

Special Requirements	Only admin accounts can update status; status must reflect correctly in customer order history.
Technology	Firestore (Orders); admin interface.
Inputs	Order selection; new status value; update action.
Outputs	1. Updated order status. 2. Confirmation/error.

Table 2.14: UC-12 Process Orders

2.8 SYSTEM SEQUENCE DIAGRAMS

System Sequence Diagrams (SSDs) represent the black-box interaction between an external actor and the System for a specific use case scenario. An SSD captures the sequence of system events (requests sent to the system) and the system's responses, without modeling internal components, database calls, or implementation logic. The SSDs below are derived from the main success scenario of each use case, with common alternative outcomes modeled using alt fragments.

2.8.1 SSD-1: Create Customer Account

Figure 2.3 shows the customer registration flow at the system level. The customer submits name, email, and password to the system. If the details are valid and the email is not registered, the system creates the account and returns a customer ID; otherwise, it rejects the request and returns an error message.

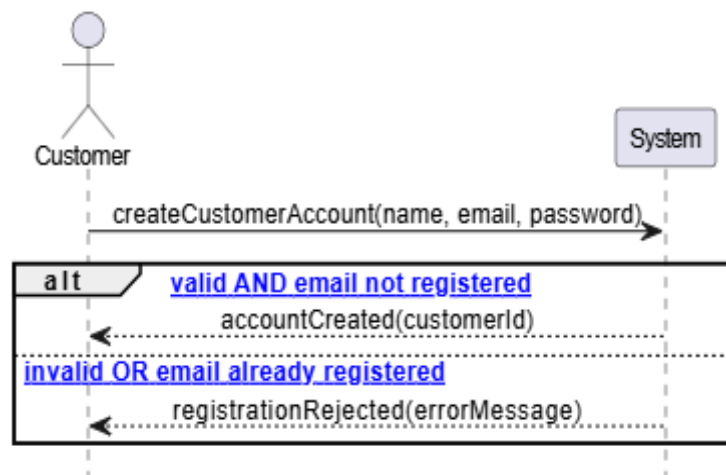


Figure 2.3: SSD-1 Create Customer Account

2.8.2 SSD-2: Explore Product Catalog

Figure 2.4 depicts how a customer requests a product catalog from the system. When a customer applies search terms or category filters, the system will deliver updated filtered results or an empty list if no matches were found.

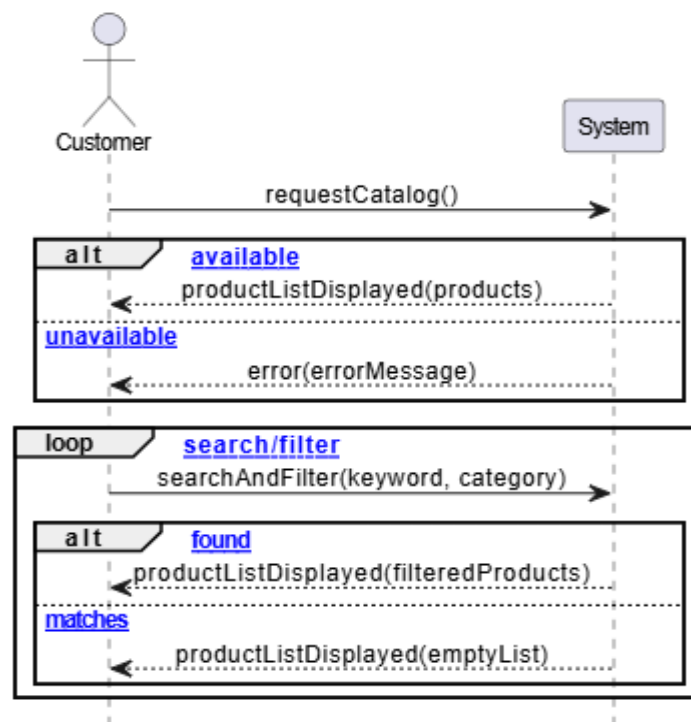


Figure 2.4: SSD-2 Explore Product Catalog

2.8.3 SSD-3: Review Product Details

Figure 2.5 illustrates how a consumer picks a product and asks for information from the system. The system displays product information, stock status, and AR availability. If the product is not available, it returns an error.

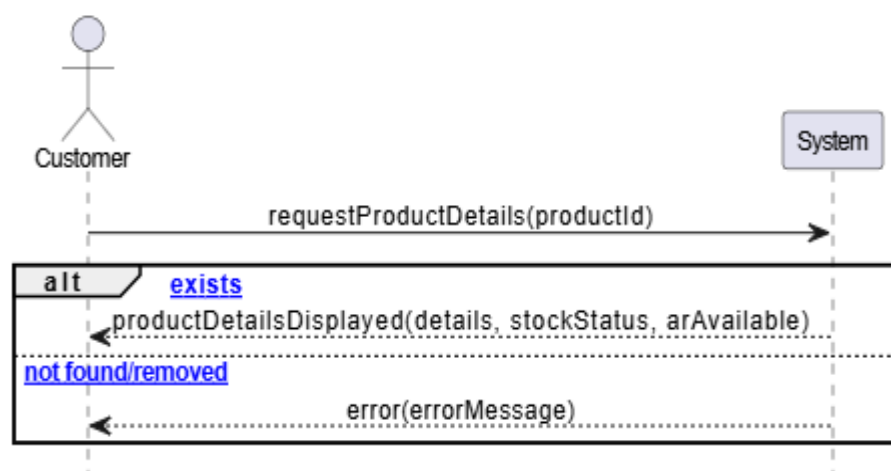


Figure 2.5: SSD-3 Review Product Details

2.8.4 SSD-4: Visualize Product in Augmented Reality

Figure 2.6 depicts a customer requesting an AR preview for a product. If AR is supported and a 3D model is available, the system will request camera permission and begin the preview. Otherwise, the system will refuse the request and provide a clear message (e.g., permission denied, unsupported device, or missing model).

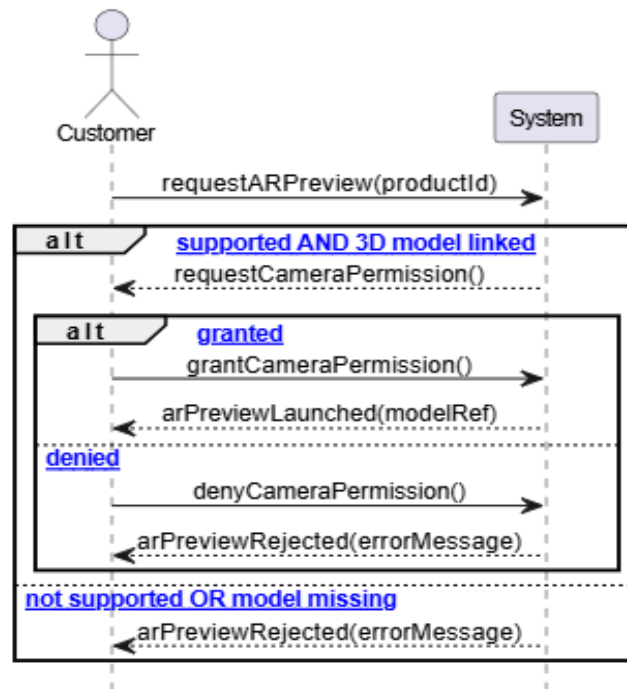


Figure 2.6: SSD-4 Visualize Product in AR

2.8.5 SSD-5: Maintain Shopping Cart

Figure 2.7 illustrates how customers can add products to their shopping, adjust quantities, or remove things. The system compares requested quantities to available stock and provides an updated cart summary or rejects the action if stock limits are exceeded.

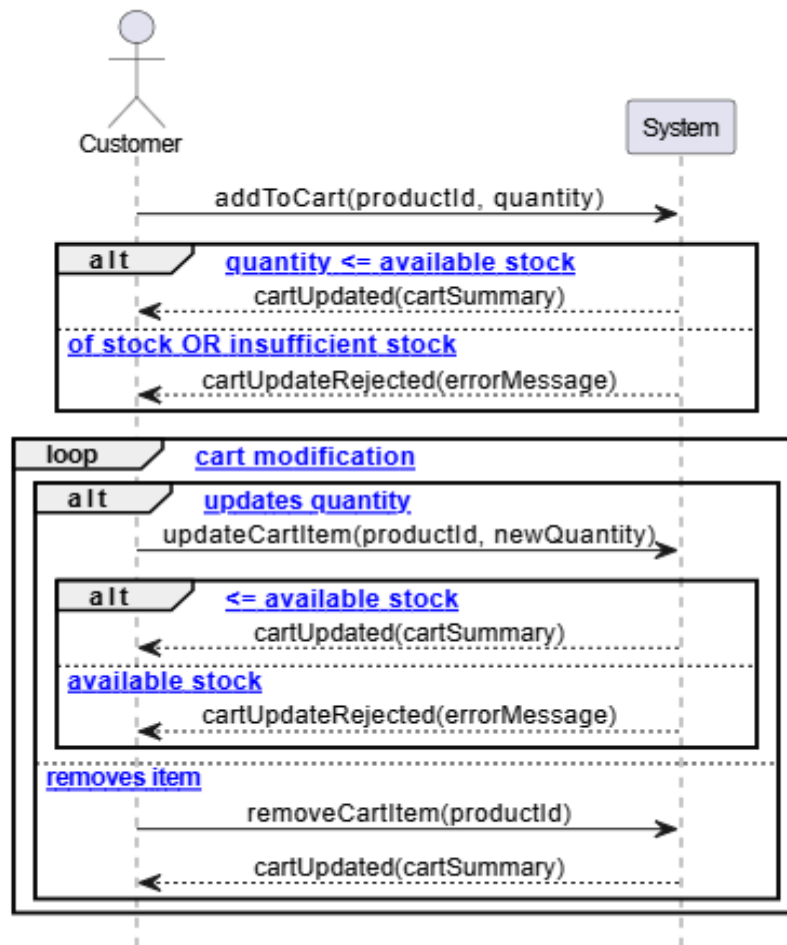


Figure 2.7: SSD-5 Maintain Shopping Cart

2.8.6 SSD-6: Checkout and Submit Order

Figure 2.8 shows the customer enters delivery details and selects a payment method to place an order. The system analyzes input and examines stock availability before confirming the order with an order ID. If validation fails or there is insufficient stock, the request is rejected.



Figure 2.8: SSD-6 Checkout and Submit Order

2.8.7 SSD-7: Review Order History

Figure 2.9 depicts a customer requesting order history from the system. The system displays the customer's orders (or an empty list if none exist) and allows them to view specific order details, including goods and status.

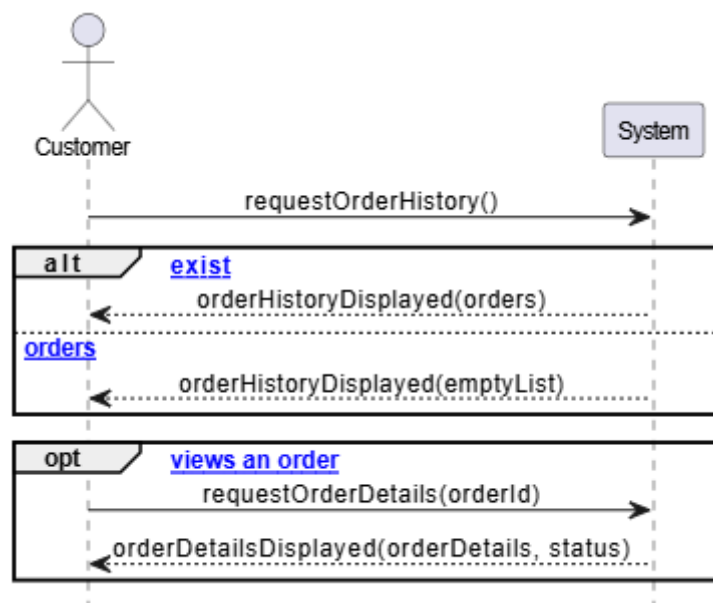


Figure 2.9: SSD-7 Review Order History

2.8.8 SSD-8: Maintain Customer Profile

Figure 2.10 shows the customer changing profile information through the system. The system validates the updated fields and either confirms the profile update or rejects it with an error message.

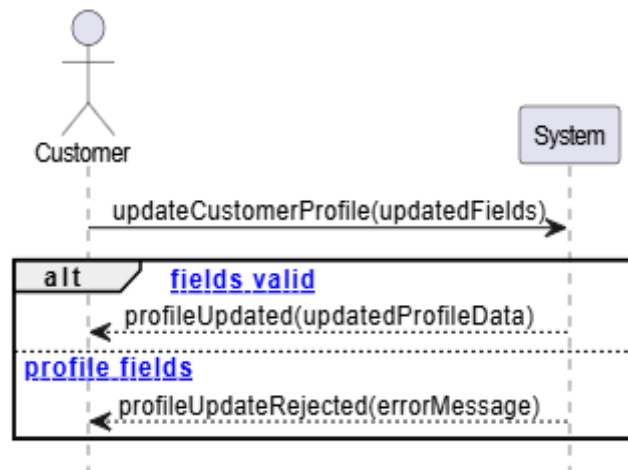


Figure 2.10: SSD-8 Maintain Customer Profile

2.8.9 SSD-9: Maintain Product Catalog

Figure 2.11 shows the administrator performing product catalog maintenance by adding, updating, or deleting products. The system validates the request and confirms success or returns an error if the operation fails.

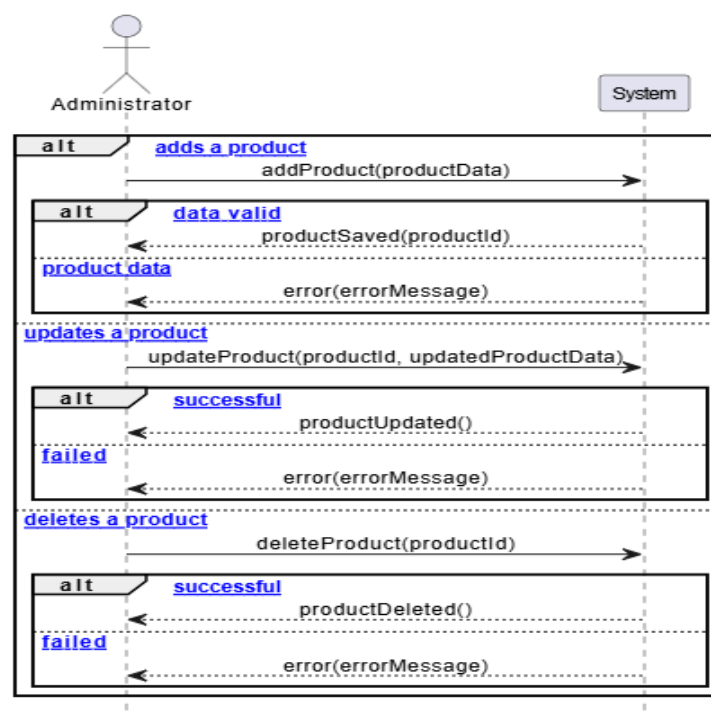


Figure 2.11: SSD-9 Maintain Product Catalog

2.8.10 SSD-10: Maintain Product Media Assets

Figure 2.12 shows the administrator uploading product images or associating a 3D model for AR preview. The system stores the media and confirms the generated URLs/model reference or returns an error if the upload or validation fails.

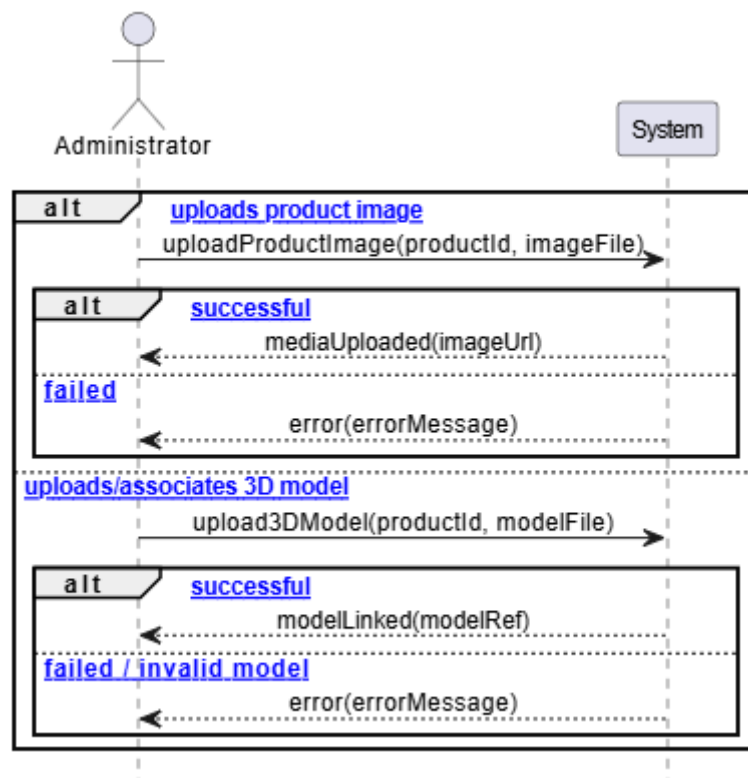


Figure 2.12: SSD-10 Maintain Product Media Assets

2.8.11 SSD-11: Maintain Inventory/Stock

Figure 2.13 shows the administrator updating stock quantity for a product. The system validates that the value is a non-negative integer and confirms the update, or rejects it if the value is invalid.

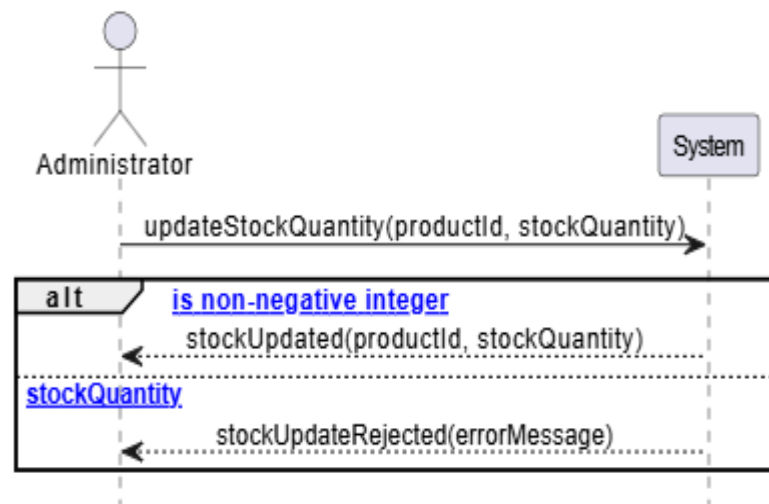


Figure 2.13: SSD-11 Maintain Inventory/Stock

2.8.12 SSD-12: Process Orders

Figure 2.14 shows the administrator updating the status of an order (e.g., pending to shipped). The system saves the updated status and confirms success or returns an error if the update fails.

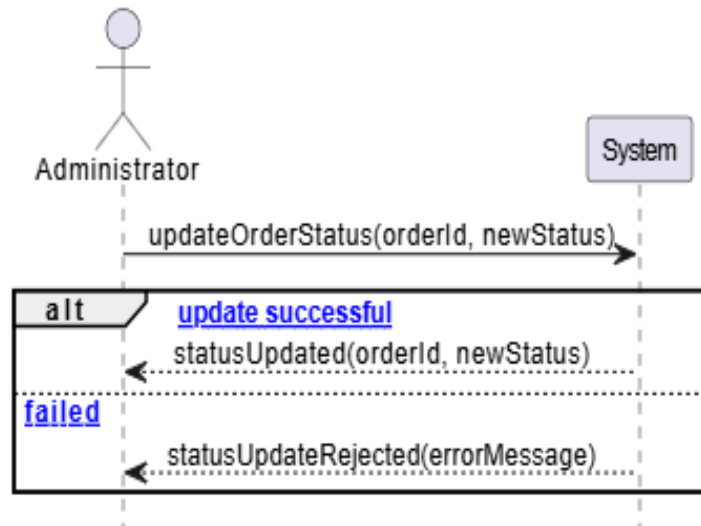


Figure 2.14: SSD-12 Process Orders

2.9 DOMAIN MODEL

A domain model is a conceptual view of the problem domain that identifies the system's core real-world entities, their key attributes, and the associations between them. It helps define what information the system must manage.

Figure 2.15 depicts the conceptual domain elements and relationships in the AR-enabled e-commerce system. It depicts how a customer manages a cart and places orders containing order goods, with each order including delivery details and optional payment information. The product catalog is organized into categories, product photos, optional AR 3D model assets, and inventory stock. The model also provides enumerations (OrderStatus, PaymentMethod, and PaymentStatus) that specify the valid states for order processing and payment.

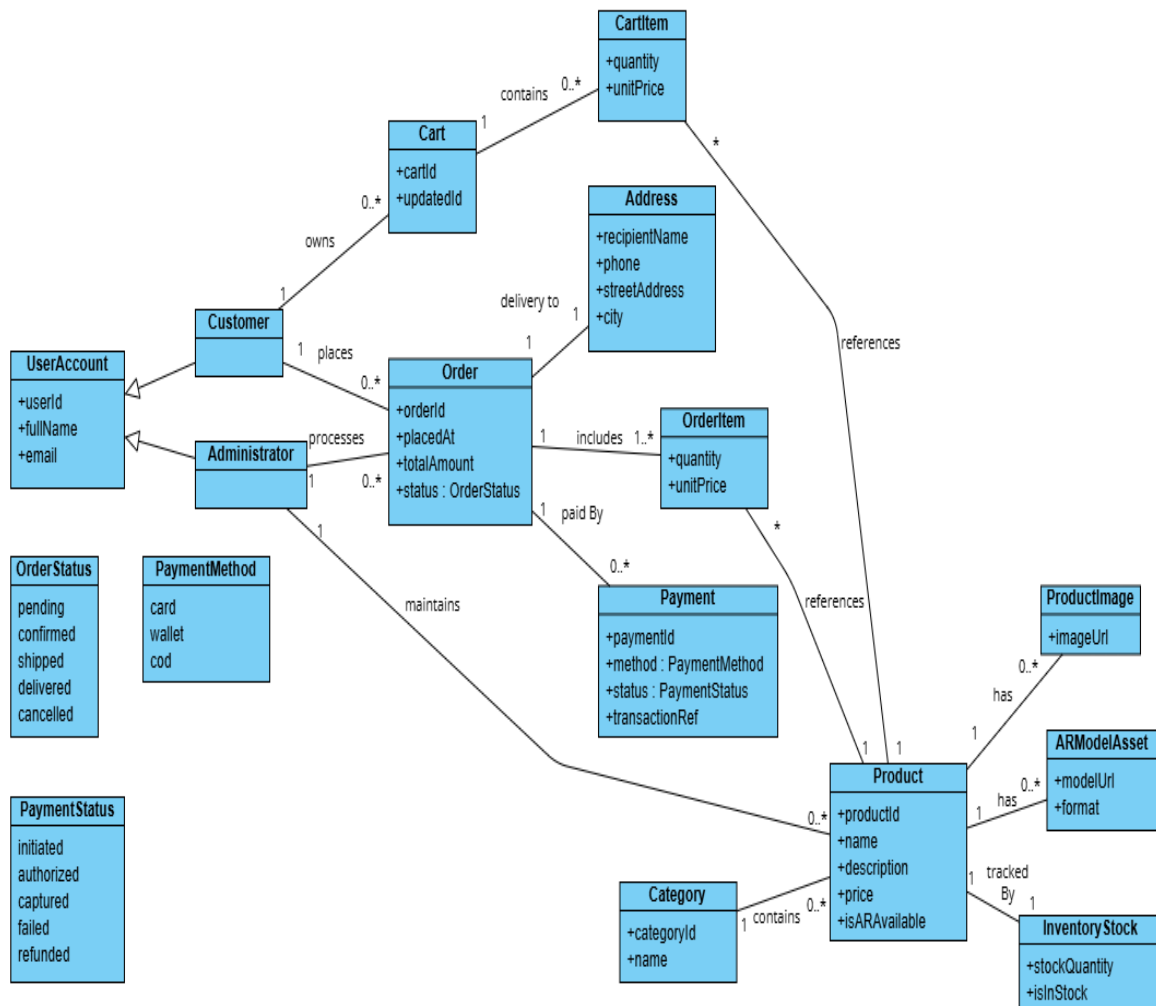


Figure 2.15: Domain Model

CHAPTER # 3
SOFTWARE DESIGN DESCRIPTION (SDD)

3.1 INTRODUCTION

This chapter outlines the Software Design Description (SDD) for the E-Commerce Platform with Augmented Reality. The document outlines the system's design approach and major components, including product browsing, AR preview, cart and checkout, and order administration. The modular design simplifies implementation, testing, and maintenance by keeping components loosely connected.

3.1.1 Design Overview

The AR-enabled e-commerce system is built using a layered, modular design. Customers can browse products, examine details, preview items in AR (if available), manage their cart, and submit orders through checkout. Administrators oversee the product catalog, media assets, stock levels, and order status updates. This split ensures system maintainability and allows for gradual development and testing.

Major elements include:

- **Customer Application (Presentation Layer)**
Provides interfaces for account creation/login, catalog browsing, search/filter, product details, AR preview, cart operations, checkout, order history, and profile management.
- **Administrator Module (Administration Layer)**
Interfaces for maintaining the product catalog, uploading and associating product media (pictures and AR 3D models), updating inventory values, and managing order status.
- **Application Logic Layer (Core Services)**
Validates user input, calculates cart totals, verifies stock during checkout, creates orders, and changes order status.
- **Data Management Layer (Repositories / Storage Access)**
Stores structured records (users, products, carts, orders, inventories) and retrieves media assets (pictures, 3D models).
- **AR Preview Subsystem**
Responsible for initializing AR sessions, handling permissions, detecting planes, and rendering 3D product models in the consumer environment.
- **Payment Handling Module**
When a payment gateway is integrated, it supports online payments. Alternatively, the checkout sequence can use COD or simulated payment confirmation in the prototype.

3.1.2 Requirements Traceability Matrix

Table 3.1 shows the Requirements Traceability Matrix (RTM). Each requirement/use case is linked to its corresponding interface or module, as well as the verification test case. This mapping ensures the design meets all requirements and can be tested and validated.

Requirement ID	Requirement Name	Interface/Module	Test Case
UC-01	Create Customer Account	Signup/Login + Authentication Module	TC-01: Registration Validation (valid/duplicate/invalid)
UC-02	Explore Product Catalog	Catalog UI + Catalog Management Module	TC-02: Catalog Load + Search/Filter Accuracy
UC-03	Review Product Details	Product Details UI + Product Information Module	TC-03: Product Detail Integrity (price/media/stock)
UC-04	Visualize Product in Augmented Reality	AR Preview UI + AR Preview Subsystem + Media Access	TC-04: AR Launch (permission/device/model cases)
UC-05	Maintain Shopping Cart	Cart UI + Cart Management Module	TC-05: Cart Operations (add/update/remove + totals)
UC-06	Checkout and Submit Order	Checkout UI + Order Management Module	TC-06: Order Creation + Stock Recheck + Confirmation
UC-07	Review Order History	Order History UI + Order Query Module	TC-07: Order History Access Control + Listing
UC-08	Maintain Customer Profile	Profile UI + User Profile Module	TC-08: Profile Update Validation + Persistence
UC-09	Maintain Product Catalog	Admin UI + Product Administration Module	TC-09: Product CRUD + Field Validation
UC-10	Maintain Product Media Assets	Admin UI + Media Asset Management Module	TC-10: Image/Model Upload + Product Linking
UC-11	Maintain Inventory/Stock	Admin UI + Inventory Management Module	TC-11: Stock Update Rules (non-negative + reflected)
UC-12	Process Orders	Admin UI + Order Processing Module	TC-12: Order Status Update + Customer Visibility

Table 3.1: Requirements Traceability Matrix

3.2 SYSTEM ARCHITECTURAL DESIGN

3.2.1 Chosen System Architecture

This project uses a Three-Tier (Layered) Architecture, which separates user interface, application logic, and data storage. It seamlessly integrates with Firebase-based systems, ensuring a modular and maintainable design.

1. **Presentation Layer (Client Layer)**

- **Technology:** Flutter (Android / iOS)
- **Responsibilities:**
 - **Customer Module:** registration/login, product browsing/search/filter, product details, AR preview, cart management, checkout, order history, and profile management.
 - **Admin Module (In-App):** product management, media upload/association (images and 3D models), inventory/stock updates, and order processing (status updates).
 - Manages user input, navigation flow, and screen-level validations.

2. **Application Layer (Business Logic Layer)**

- **Technology:** Flutter service modules (app-side logic)
- **Responsibilities:**
 - Apply business rules for form validation, cart total computation, and checkout validation.
 - Verifies stock availability at checkout time.
 - Creates orders and updates order status based on the defined workflow.
 - Ensures data consistency during order placement by using Firestore Transactions so that stock validation and stock decrement occur atomically.

3. **Data Layer (Persistent Storage Layer)**

- **Technology:** Firebase Cloud Firestore + Firebase Storage
- **Responsibilities:**
 - **Cloud Firestore:** stores users, products, categories, carts, orders, and inventory values (stockQuantity).
 - **Firebase Storage:** stores product images and AR 3D model assets linked to products.
 - Uses access control through Firebase Security Rules to restrict admin-only operations (catalog/inventory/order management) and protect user-specific data (carts/orders).

3.2.2 Discussion of Alternative Designs

- **Two-Tier Architecture**

In a two-tier architecture, the majority of the logic is handled directly by the client, which communicates tightly with the database/storage layer. It was not chosen because:

- It increases complexity in enforcing consistent business rules, especially for inventory updates and order consistency.
- As features grow (AR + commerce + admin), the client becomes harder to maintain without clear layering.
- Security and access control become more difficult to manage cleanly.

- **Monolithic Custom Backend Architecture**

A monolithic backend (custom server and database) was also investigated. It was not chosen because:

- It increases hosting and maintenance overhead (API development, deployment, scaling).
- Firebase already provides authentication, structured storage, and media storage required for this project.
- The project scope is better served by focusing on core features (AR preview, shopping flow, admin management) rather than building server infrastructure.

3.2.3 System Interface Description

The system integrates via the following interfaces:

Customer Interface (Mobile UI)

- Allows customers to register/login, browse products, view product details, preview items in AR, manage cart, checkout, and view order history/profile.
- Supports interaction through forms, buttons, lists, and confirmation screens.

Admin Interface (In-App Admin Module)

- Allows administrators to add/update/delete products, upload/associate images and 3D models, update stockQuantity, and update order status.
- Access is restricted to authorized admin accounts.

Authentication Interface

- Manages account registration and login sessions.
- Ensures protected operations require authentication, and admin actions require admin authorization.

Database Interface (Cloud Firestore Access)

- Stores and retrieves structured data including products, categories, carts, orders, user profiles, inventory, and order statuses.
- Supports query-based access for catalog listing, filtering, and customer-specific order history.

Media Storage Interface (Firebase Storage Access)

- Stores product images and AR 3D assets.
- Provides secure references/URLs used by the application to load images and render AR models.

AR Framework Interface

- Creates AR sessions, requests camera permission, detects surfaces, and renders the selected product's 3D model.
- Handles AR failure cases such as unsupported device, permission denial, and tracking/lighting limitations.

Payment Gateway Interface

- Supports online payment authorization/capture when implemented.
- Returns payment success/failure status to complete or reject the checkout flow accordingly.

3.3 DETAILED DESCRIPTION OF COMPONENTS

This section outlines the essential components of the E-Commerce Platform with Augmented Reality using the chosen layered architecture. The system is organized into presentation, application logic, and data/storage responsibilities. Each component is intended to meet a specific set of needs and interacts with the others to facilitate product browsing, AR preview, cart and checkout, inventory management, and order processing.

3.3.1 User Interface (Presentation Layer)

The user interface controls all user-system interactions via the mobile application. It includes the screens and routines required for both the customer and administrator roles. Customer experience comprises signup/login, catalog browsing, product information, AR preview, cart, checkout, order history, and profile displays. Administrators can access screens for managing products, uploading media, updating inventories, and updating order status. The UI provides consistent navigation, form validation, and unambiguous feedback on success/failure stages.

3.3.2 Controller and State Management Component

This component coordinates UI displays and underlying services. The system processes user actions (e.g. Add to Cart, Checkout, Upload Model) and modifies the user interface accordingly. It manages screen transitions, loading/error states, and synchronizes the UI with current system data (stock status, cart totals, and order status changes).

3.3.3 Authentication and Authorization Component

The authentication component manages user registration, login, and sessions. Management using the authenticating service. The authorization section restricts consumer features to authenticated customers. Administrative functions are only available to approved admin accounts. This component prevents unauthorized access to protected actions such as the cart, checkout, purchase history, and admin pages.

3.3.4 Catalog and Product Management Component

This component oversees the entire product catalog lifetime. It fetches and displays product categories and lists, allows for search and filtering, and delivers comprehensive product information upon selection. This assures continuous availability of essential product attributes for the app, including pricing, pictures, stock quantity, and AR model references (if relevant). This component handles administrative product CRUD operations (add, update, and delete) and validates needed values.

3.3.5 AR Preview Component

The AR preview component enables display of supported products and devices. The AR session starts, camera permissions are checked, the connected 3D model asset is loaded, and the model is rendered on recognized surfaces. Guided feedback is provided when AR cannot be initiated due to permission refused, unsupported device, model load failure, or inadequate tracking/surface detection. This allows users to safely return to product details without experiencing program instability.

3.3.6 Cart Management Component

The cart component handles user-specific cart activities such as add, remove, and quantity change. It recalculates the subtotal/total and enforces stock limitations to ensure that the requested quantities do not exceed the stock quantity. Cart state is saved per user to ensure continuity across sessions and devices, and it is consistent with the product catalog (out-of-stock handling and quantity limitations).

3.3.7 Checkout, Order Processing, and Payment Component

This component handles checkout, delivery validation, and payment processing for online transactions. To ensure consistency, order confirmation triggers atomic order placement using Firestore Transactions. This involves rechecking stock, creating the order, decrementing stock quantity for ordered products, and clearing the cart in a single transaction. It allows for administrative order processing, including status transitions from pending to shipped to delivered, and updates the customer's purchase history accordingly.

3.3.8 Data Access and Media Storage Component (Data Layer)

This component handles all interactions with persistent storage. The database service stores and retrieves structured records such as users, products, carts, orders, and inventory quantities. Product photos and 3D model assets are saved in media storage and linked via references/URLs. Access control is enforced using security rules, limiting consumer access to their own cart/orders and restricting catalog/inventory/order adjustments to administrators.

3.4 ARCHITECTURAL DIAGRAM

Figure 3.1 depicts the three-tier architecture of an AR-enabled e-commerce system. The Flutter app has three layers: the Presentation Layer (customer and admin interfaces), the Application Layer (controller/state management and service modules), and the Data Layer (Firebase services, ARCore/ARKit, and external payment gateway).

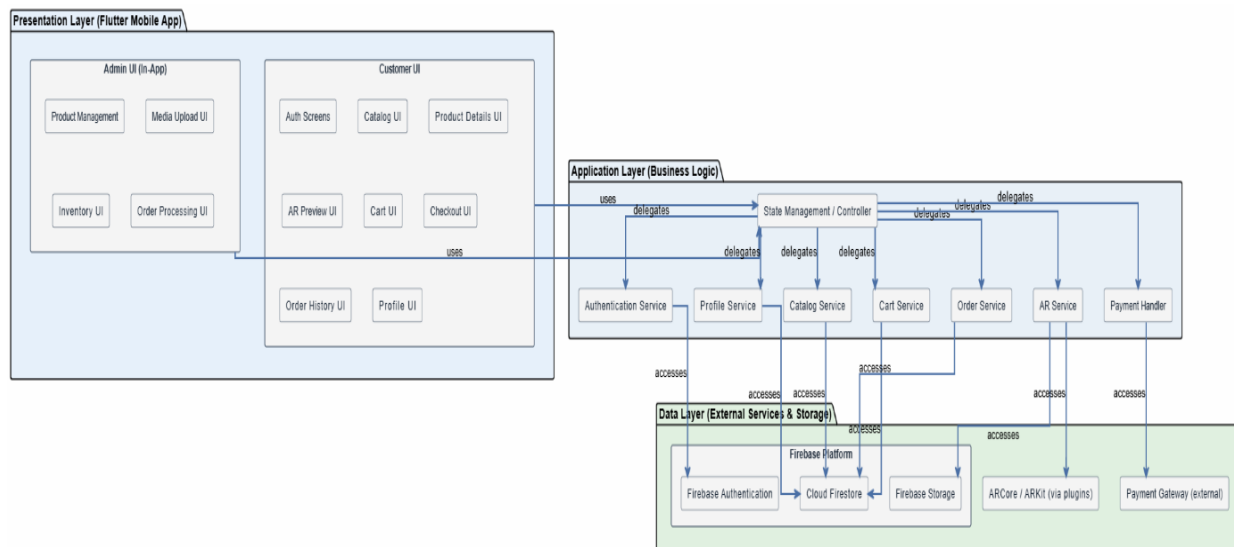


Figure 3.1: Three-Tier Layered Architecture Diagram

3.5 SEQUENCE DIAGRAMS

Sequence diagrams depict how actors and internal components communicate over time to complete a certain use case. Unlike system sequence diagrams (SSDs), these diagrams show collaboration across UI displays, controllers/services, domain objects, and external platforms.

3.5.1 SD-1: Create Customer Account

Figure 3.2 depicts the customer registration flow, with the Signup/Login UI sending information to the AuthController/AuthService. The solution creates a user using Firebase Authentication and stores their profile in Cloud Firestore upon success. Otherwise, it produces an error and displays a failure notice on the user interface.

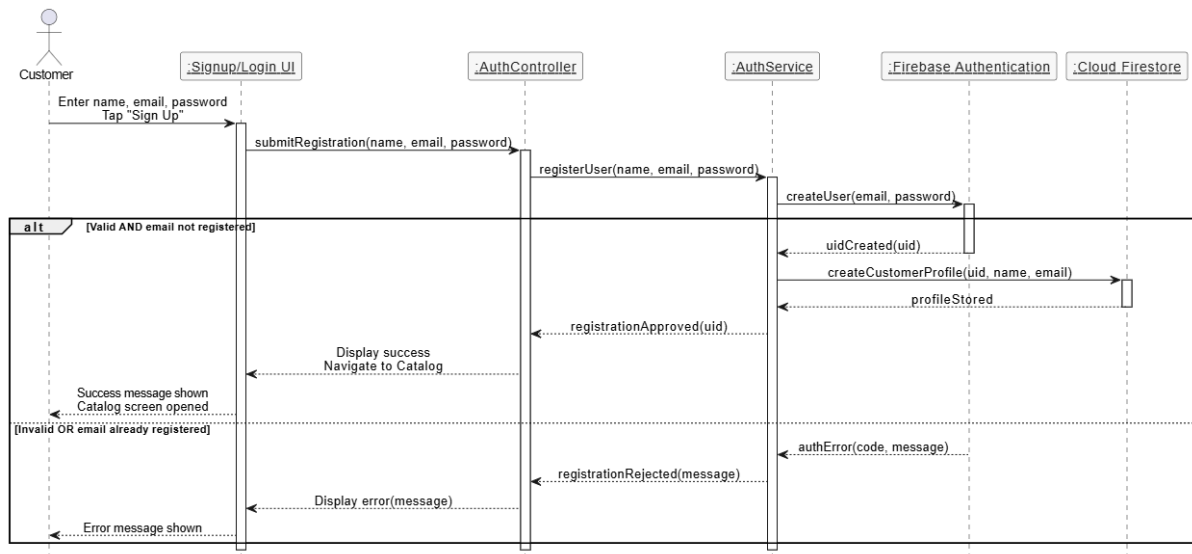


Figure 3.2: SD-01 Create Customer Account

3.5.2 SD-2: Explore Product Catalog

Figure 3.3 depicts how the Customer UI requests a product catalog from the Catalog Service, which retrieves product records from Cloud Firestore and provides them for rendering. If the request fails, the UI will display an error message and allow you to retry.

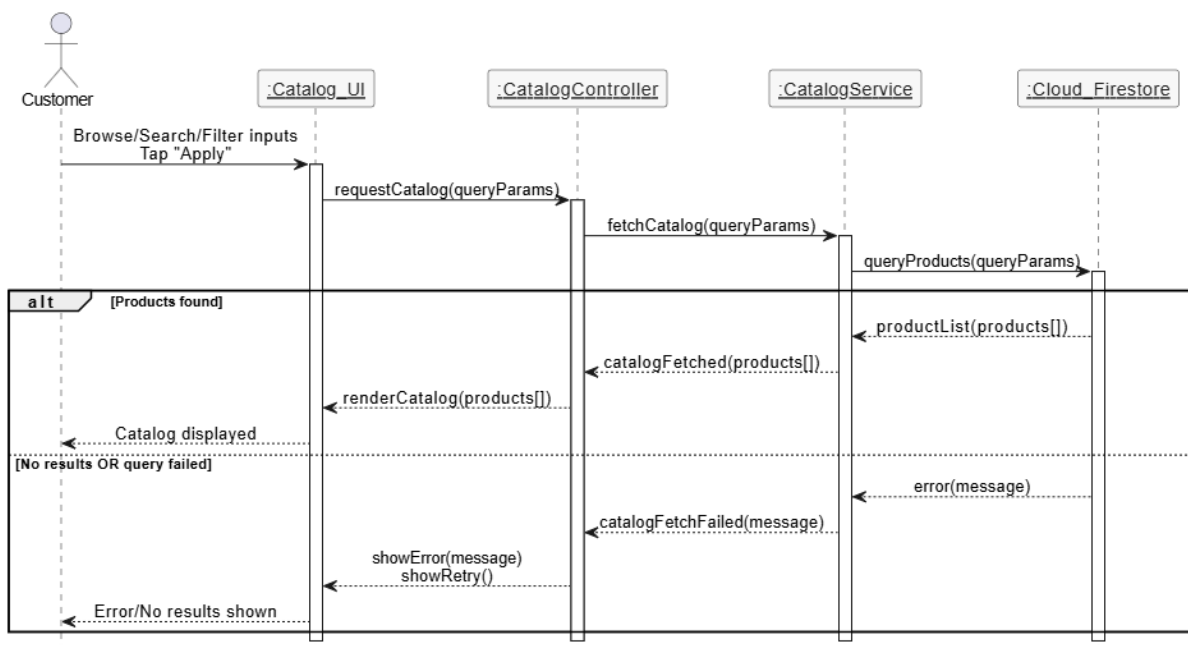


Figure 3.3: SD-02 Explore Product Catalog

3.5.3 SD-3: Review Product Details

Figure 3.4 displays the Product Details UI, which retrieves complete product information from Cloud Firestore using the Catalog Service. This includes price, description, media links, stock indicator, and AR availability.

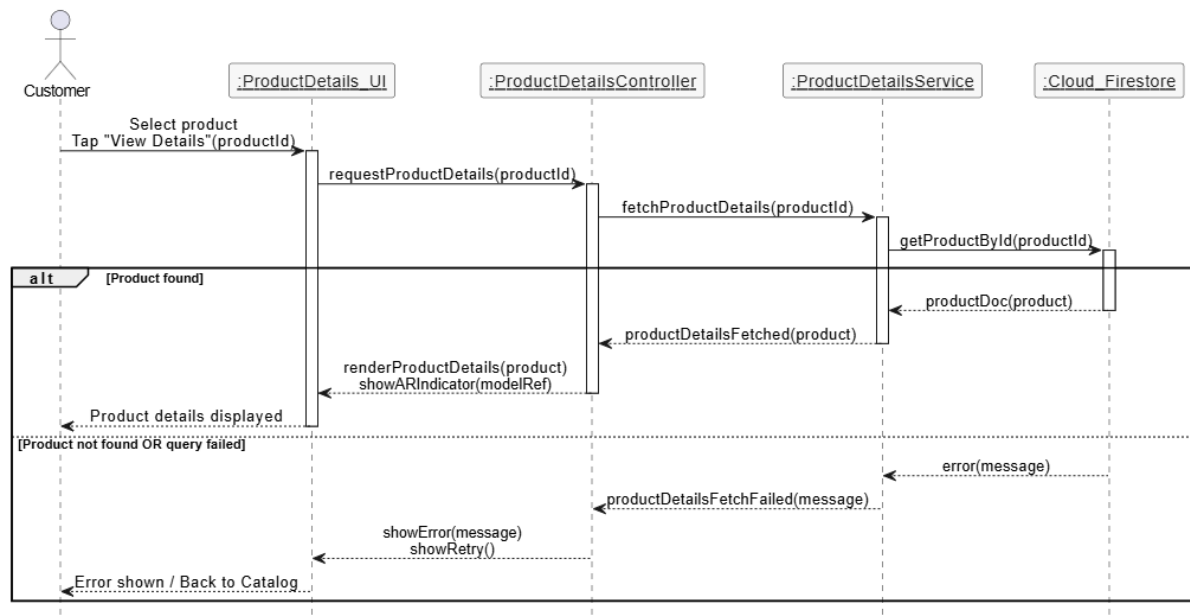


Figure 3.4: SD-03 Review Product Details

3.5.4 SD-4: Visualize Product in Augmented Reality

Figure 3.5 shows the AR Preview UI starts an AR session using the AR Service. The product's 3D model is loaded from Firebase Storage and rendered using ARCore/ARKit. Alternative paths address issues such as lacking permissions, unsupported devices, and missing models.

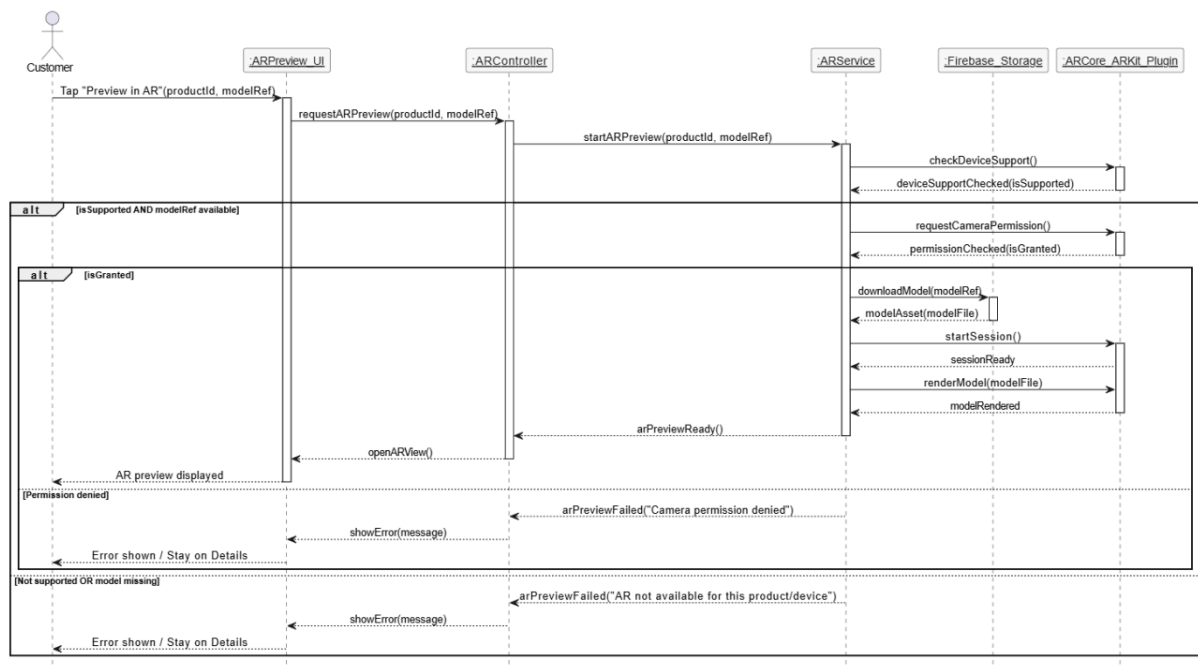


Figure 3.5: SD-04 Visualize Product in AR

3.5.5 SD-5: Maintain Shopping Cart

Figure 3.6 illustrates how cart operations (add/update/remove) are sent from the Customer UI to the Cart Service, which changes the user's cart state in Cloud Firestore and displays refreshed totals in the UI.

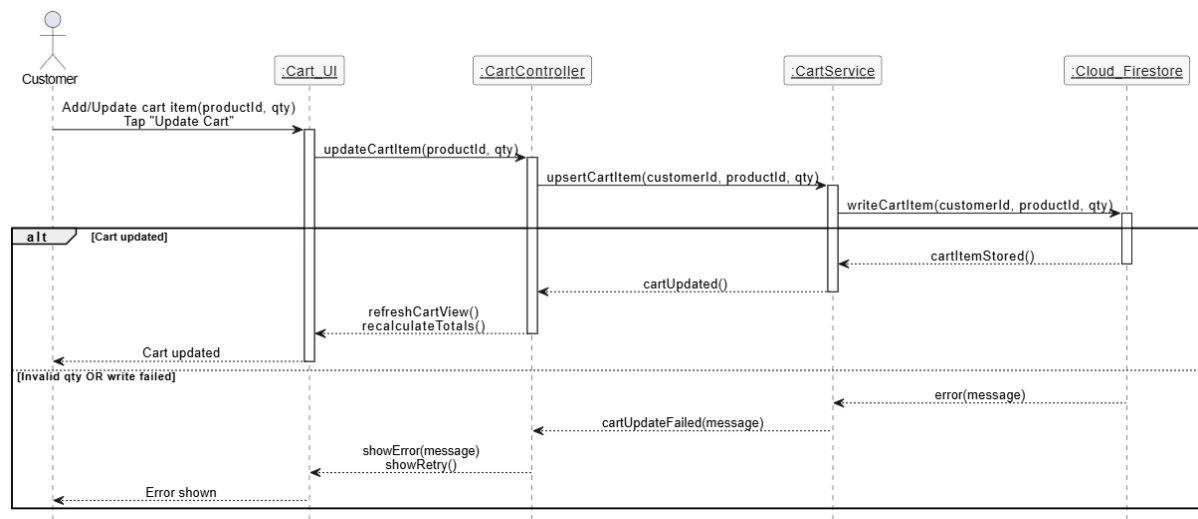


Figure 3.6: SD-05 Maintain Shopping Cart

3.5.6 SD-6: Checkout and Submit Order

Figure 3.7 shows the Checkout UI uses a Firestore transaction to place an order (stock validation, order generation, stock decrement, and cart clearing), while the Payment Handler communicates with the Payment Gateway for online payment.

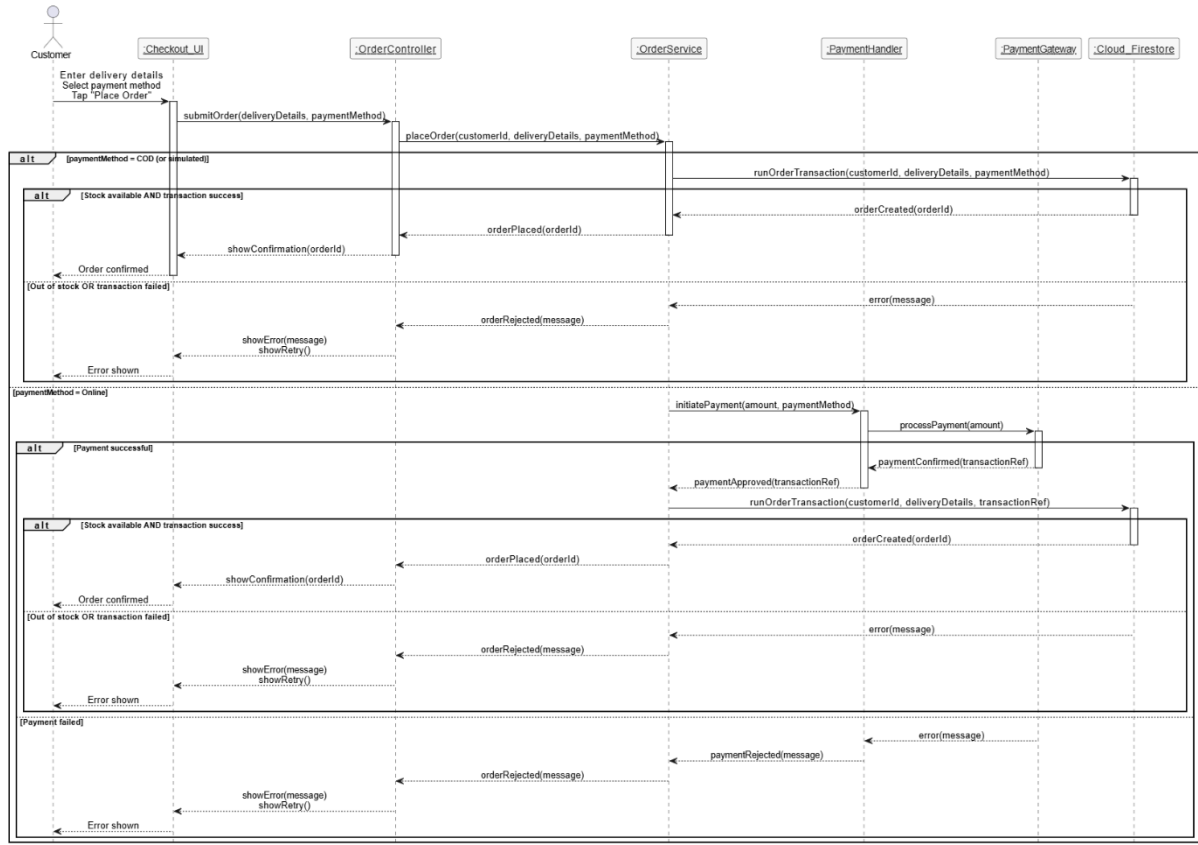


Figure 3.7: SD-06 Checkout and Submit Order

3.5.7 SD-7: Review Order History

Figure 3.8 shows the Order History UI requests customer orders from the Order Service, which queries Cloud Firestore with the authenticated user context and displays the list.

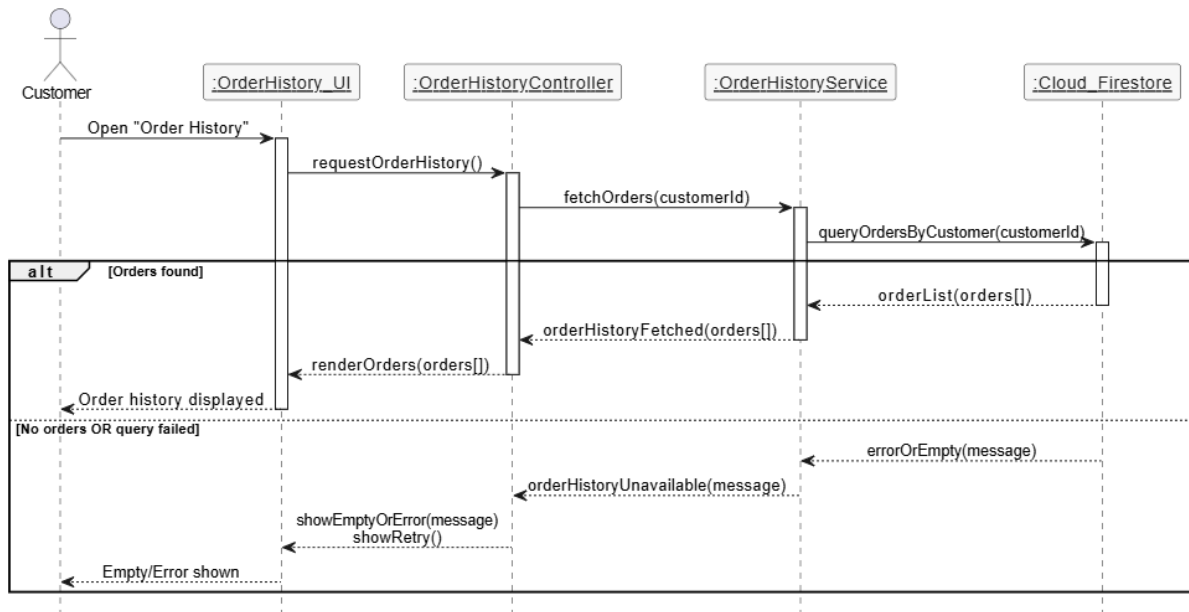


Figure 3.8: SD-07 Review Order History

3.5.8 SD-8: Maintain Customer Profile

Figure 3.9 shows the Profile UI submits profile adjustments to the Profile Service, which validates and stores the changes in Cloud Firestore and provides a success/error response to the UI.

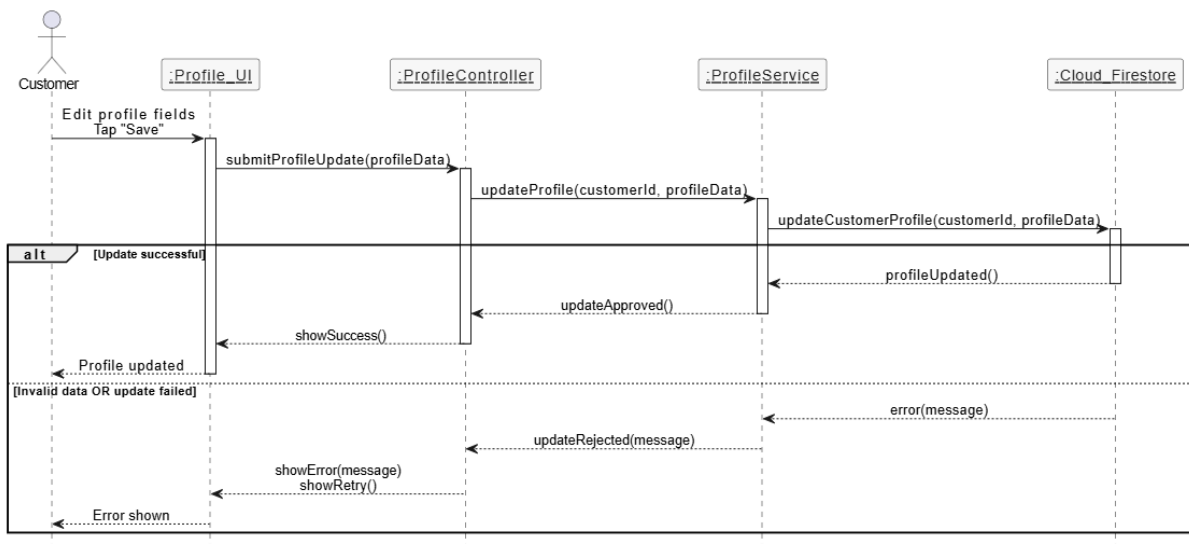


Figure 3.9: SD-08 Maintain Customer Profile

3.5.9 SD-9: Maintain Product Catalog

Figure 3.10 depicts the Admin UI for creating, updating, and deleting products using the Admin Management Service, which updates product records in the Cloud Firestore and return the operation status.

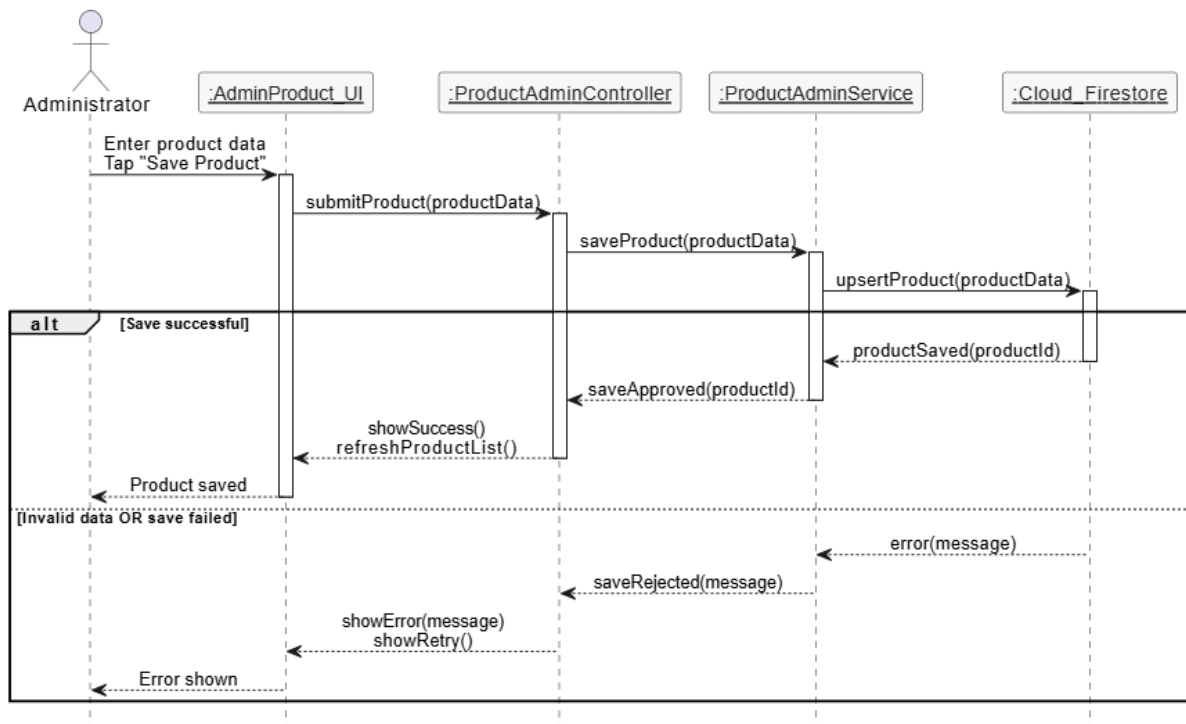


Figure 3.10: SD-09 Maintain Product Catalog

3.5.10 SD-10: Maintain Product Media Assets

Figure 3.11 shows the admin uploads product images/3D models using the Admin UI. The Media Handler stores the files in Firebase Storage, gets URLs, and updates the product document in Cloud Firestore.

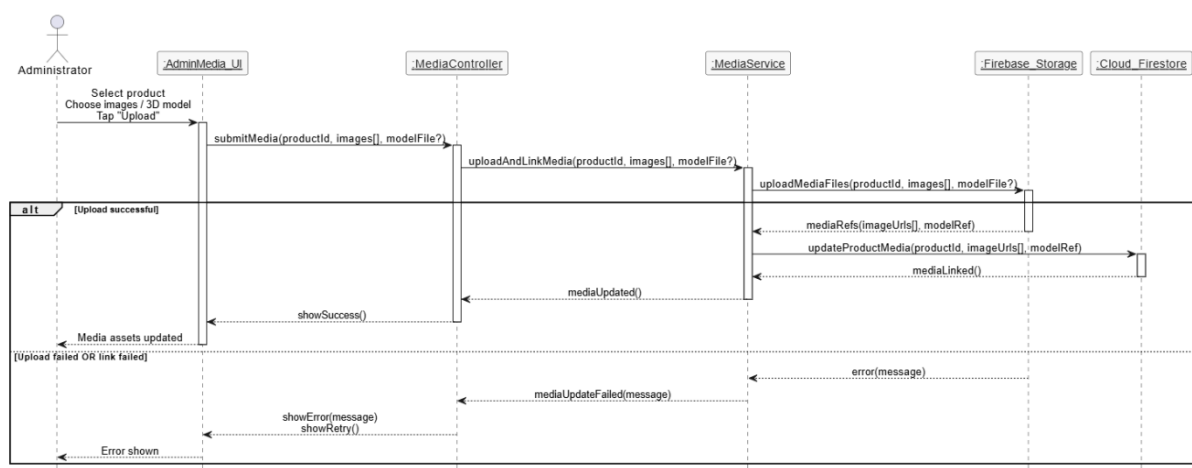


Figure 3.11: SD-10 Maintain Product Media Assets

3.5.11 SD-11: Maintain Inventory/Stock

Figure 3.12 shows the admin updates stock amounts from the Inventory UI using the Admin/Inventory Service. The changed fields are then written to Cloud Firestore and validated for non-negative quantities.

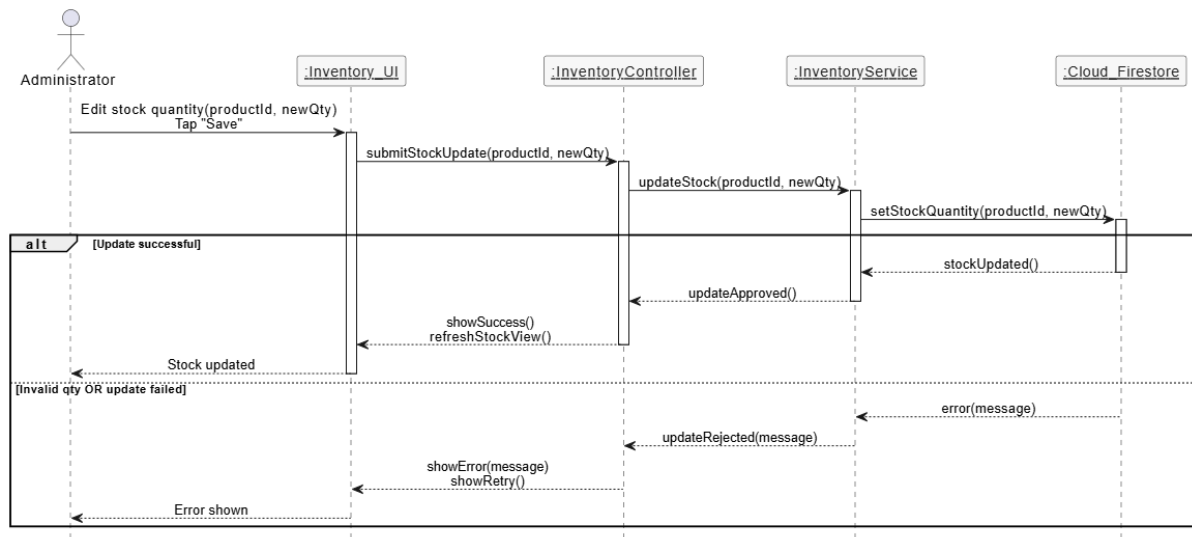


Figure 3.12: SD-11 Maintain Inventory/Stock

3.5.12 SD-12: Process Orders

Figure 3.13 shows the admin retrieves orders and adjusts their status using the Order Management UI and Order Service. This updates the order status fields in Cloud Firestore, allowing customers to view the most recent tracking state.

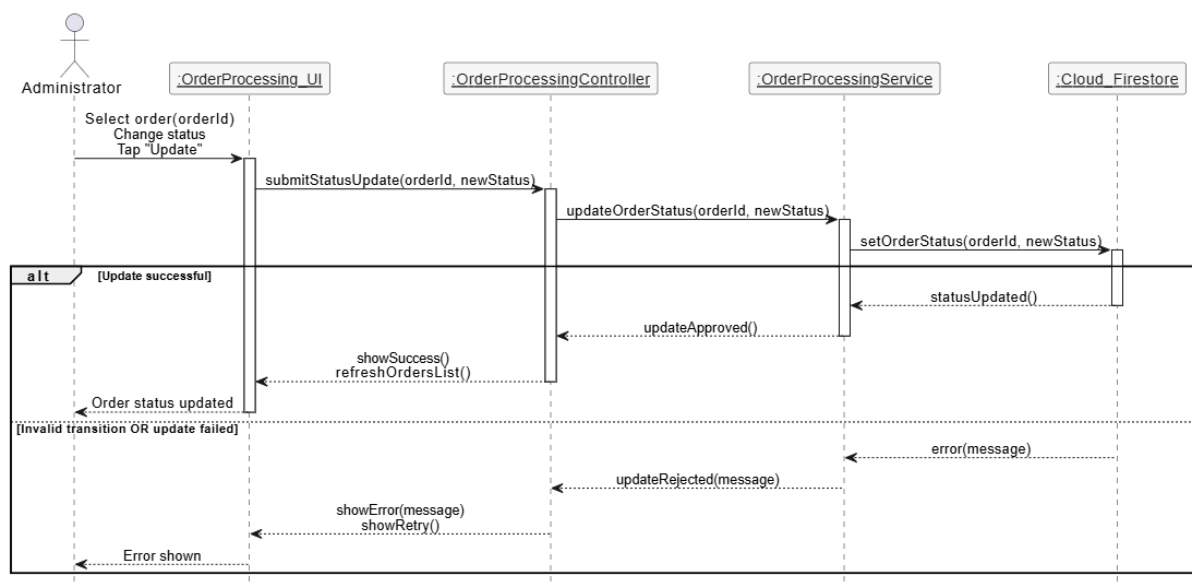


Figure 3.13: SD-12 Process Orders

3.6 CLASS DIAGRAM

Figure 3.14 depicts the class structure of the AR-enabled e-commerce system, which includes the key classes, attributes/operations, and relationships. User Account is divided into two categories: customer and administrator. A customer owns a Cart (which contains Cart Item entries) and places Orders (which contain Order Item entries) that are delivered to an address. Each Order may have an associated Payment, whose behavior is determined by the Payment Method and Payment Status enumerations, whereas Order Status determines the order's lifespan. Products are categorized by Category, enhanced with Product Image and optional AR Model Asset for AR preview, and availability is managed using Inventory Stock.

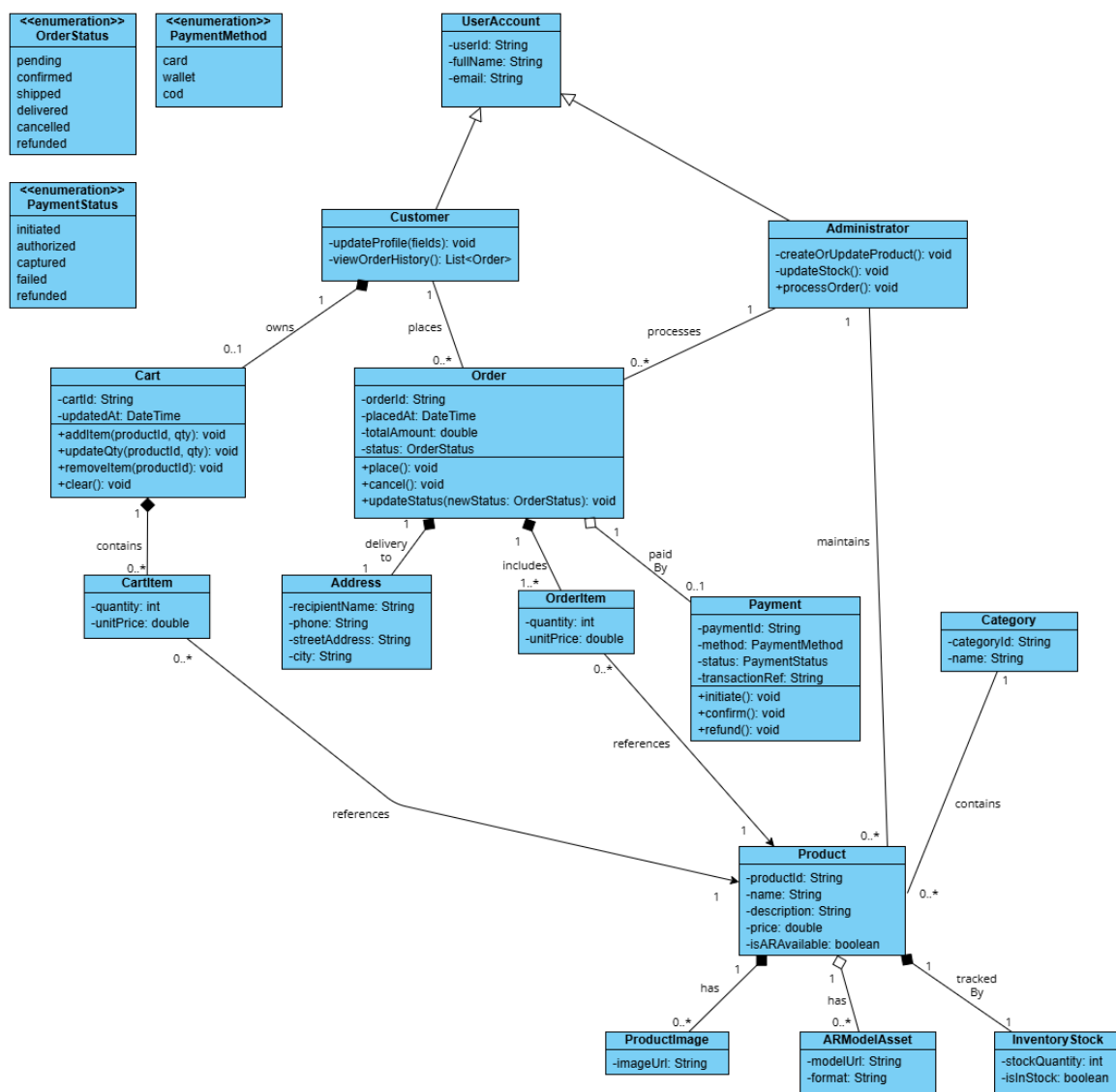


Figure 3.14: Class Diagram

CHAPTER # 4

SOFTWARE TEST DOCUMENTATION (STD)

4.1 INTRODUCTION

This chapter outlines the testing plan for the E-Commerce Platform with Augmented Reality to ensure the application's reliability, accuracy, and usability before final deployment. Testing involves checking the entire shopping cycle, including authentication, catalog, product information, cart, checkout, and order history. It also validates AR preview behavior on supported devices and ensures data integrity and access control across Firebase services.

4.1.1 System Overview

The suggested system is a Flutter-based mobile app with two roles: customer and administrator. Customers can register, browse, search, check product details, use AR previews, manage carts, place orders, and examine order history. Administrators control the product catalog, pictures, and 3D models, update stock, and update order statuses.

The system utilizes Firebase Authentication for user accounts, Cloud Firestore for app data, and Firebase Storage for product photos and 3D assets. The Flutter AR library supports ARCore/ARKit for AR preview, and payments can be made via an online gateway or COD/simulated payment if necessary.

4.1.2 Test Approach

Testing will be done utilizing both automated and manual approaches. Automated tests will cover basic logic (validation, cart totals, stock updates, order status changes), Firebase integration (auth and database operations), and important user flows like signup, browsing, checkout, and order history. Manual testing will evaluate AR preview functioning on supported devices, as well as usability and error handling. Non-functional checks will include screen loading and response time, as well as security measures like role-based access and user data separation.

4.2 TEST PLAN

4.2.1 Features to be Tested

The following aspects of the AR-enabled e-commerce system will be evaluated:

- Customer Registration and Login.
- Browse Product Catalog (including Search and Filter).
- View Product Details (including AR availability).
- AR Product Preview (device supported + model available cases).
- Shopping Cart Management (add/update/remove items, totals).
- Checkout and Place Order (validation + order creation).
- View Order History and Order Status.

- Admin Product Management (add/update/delete products).
- Admin Media Management (upload images and 3D models).
- Admin Inventory/Stock Updates.
- Administrative Order Processing (see orders and adjust status).

4.2.2 Features not to be Tested

The following elements are not within the scope of testing for this phase:

- AR preview on devices that do not support ARCore/ARKit.
- Highly realistic rendering checks (perfect lighting/shadows). Testing will focus on correct placement, scaling behavior, and stability.
- Third-party payment gateway certification/testing (PCI-level validation). If payment integration is incomplete, checkout will be validated using COD/simulated payment.

4.2.3 Testing Tools and Environment

Testing Tools:

- Flutter test for unit and widget tests.
- Flutter integration test for end-to-end flows on emulator/device.
- Firebase Emulator Suite or a dedicated Firebase test project for safe database/auth testing.
- Android Studio tools for device logs and debugging during AR testing.

Testing Environment:

- **Hardware:** Android/IOS smartphone (AR-supported for AR tests). Laptop/Desktop for development and admin testing.
- **OS:** Windows 10/11 (development), Android / IOS (runtime testing).
- **Framework/Platform:** Flutter (Dart), Firebase (Auth, Firestore, Storage).
- **Network:** Wi-Fi / 4G for catalog and media loading tests.

4.3 TEST CASES

4.3.1 TC-1: Registration Validation (Valid/Duplicate/Invalid)

Table 4.1 shows the test case for customer registration. The system only creates accounts with proper details and rejects erroneous inputs including missing fields, incorrect email format, weak passwords, or duplicate emails. It also ensures clear feedback is provided to the user.

Test Case: TC-1	Registration Validation (Valid/Duplicate/Invalid)
Purpose	Verify that account creation accepts valid data and rejects invalid or duplicate credentials.
Inputs	Valid: name, email (new), password (meet rules). Invalid: missing fields/weak password/invalid email. Duplicate: already-registered email.
Pass/Fail Criteria	<u>Pass</u> : account created for valid data. Meaningful error shown for invalid/duplicate data. <u>Fail</u> : invalid data accepted, or valid data rejected without clear feedback.
Expected Outputs	<u>Success</u> : customer account created and profile initialized. <u>Failure</u> : error message displayed and no account created.
Test Procedure	1. Open Sign-Up screen. 2. Provide accurate information and verify account creation. 3. Check for invalid details and validation failures. 4. Submit duplicate emails and confirm duplication handling.

Table 4.1: TC-01 Registration Validation (Valid/Duplicate/Invalid)

4.3.2 TC-2: Catalog Load + Search/Filter Accuracy

Table 4.2 shows a test example for catalog loading and search/filter behavior. Validation ensures accurate product list retrieval and display, as well as relevant search results from keywords and categories. It also addresses scenarios of no results and query failures, with appropriate UI responses.

Test Case: TC-2	Catalog Load + Search/Filter Accuracy
Purpose	Verify catalog loading and that search/filter return correct product results.
Inputs	Search keyword; category filter; empty keyword; combination of keyword + category.
Pass/Fail Criteria	<u>Pass</u> : correct products displayed for each query/filter. (no results) state appears when applicable. <u>Fail</u> : incorrect/irrelevant results or app fails to update list.
Expected Outputs	<u>Success</u> : Catalog loads and displays products that match the applied search/filter. (No results found) appear if the query returns to the empty results. <u>Failure</u> : An error message is shown and a retry option is available. No incorrect products are displayed.
Test Procedure	1. Open Catalog screen. 2. Verify initial product list loads. 3. Apply keyword search and verify results. 4. Apply category filter and verify results. 5. Apply both and verify results/empty state.

Table 4.2: TC-02 Catalog Load + Search/Filter Accuracy

4.3.3 **TC-3: Product Detail Integrity (Price/Media/Stock/AR Flag)**

Table 4.3 presents the test case for product detail correctness. It ensures that product information such as name, description, price, images, stock availability, and AR indicator matches the stored record. It also verifies that the AR option is only shown/enabled when the product has a valid 3D model reference.

Test Case: TC-3	Product Detail Integrity (Price/Media/Stock/AR Flag)
Purpose	Verify that product details show correct data including stock status and AR availability indicator.
Inputs	Select product (in-stock with images). Select product (out-of-stock). Select product (AR model available). Select product (no AR model).
Pass/Fail Criteria	<u>Pass:</u> details match stored data. Stock shown correctly. AR option enabled only when model exists (and device supports AR). <u>Fail:</u> mismatch in price/stock/media, or AR shown incorrectly.
Expected Outputs	<u>Success:</u> Product details show correct name, price, description, images, stock state, and AR availability indicator (AR enabled only when model exists). <u>Failure:</u> A clear error message is shown (failed to load details), and the user can retry or return to catalog.
Test Procedure	1. Open a product from catalog. 2. Verify price/description/images. 3. Verify stock indicator for in-stock/out-of-stock products. 4. Verify AR indicator for products with/without model reference.

Table 4.3: TC-03 Product Detail Integrity (Price/Media/Stock/AR Flag)

4.3.4 **TC-4: AR Preview Launch (Permission/Device/Model Cases)**

Table 4.4 presents the test case for AR preview. It confirms that AR launches on supported devices when permission is granted and the 3D model is available. It also verifies safe handling of failure scenarios such as permission denial, unsupported devices, or model loading errors without crashing the app.

Test Case: TC-4	AR Preview Launch (Permission/Device/Model Cases)
Purpose	Verify AR preview behavior for supported devices and proper handling of failure conditions.

Inputs	Product with valid 3D model. Camera permission (granted/denied). AR-supported vs non-supported device.
Pass/Fail Criteria	<u>Pass:</u> AR launches with model placement on supported device with permission. Clear error shown for denied permission/unsupported device/model load failure. <u>Fail:</u> crash, blank view, or misleading AR availability.
Expected Outputs	<u>Success:</u> AR view opens, camera feed starts, and the product model loads and can be placed in the environment. <u>Failure:</u> A clear message is shown for permission denied/unsupported device/model load failure and the user returns safely to product details.
Test Procedure	1. Select an AR-enabled product. 2. Tap (View in AR). 3. Test permission granted path and verify model placement. 4. Deny permission and verify message + safe return. 5. Repeat on unsupported device and verify AR blocked with reason.

Table 4.4: TC-04 AR Preview Launch (Permission/Device/Model Cases)

4.3.5 TC-5: Cart Operations (Add/Update/Remove + Totals + Stock Limit)

Table 4.5 shows a test case for shopping cart activities. It ensures accuracy when adding, updating, removing, and recalculating totals. It enforces stock restrictions and notifies users when requested quantities exceed available supply.

Test Case: TC-5	Cart Operations
Purpose	Verify cart behavior, correct totals, and enforcement of stockQuantity limits.
Inputs	Add product to cart. Update quantity (within stock). Update quantity (exceeds stock). Remove item.
Pass/Fail Criteria	<u>Pass:</u> cart updates persist. Totals correct. Quantity cannot exceed stock. <u>Fail:</u> wrong totals, quantity exceeds stock, or cart state not saved.
Expected Outputs	<u>Success:</u> Cart items are added, modified, or withdrawn correctly, and totals are accurate. Quantity is restricted to available supply. If stock is exceeded or an operation fails, the cart is not wrongly updated and an error/stock notice is displayed. <u>Failure:</u> If stock is exceeded or operation fails, the cart is not updated incorrectly, and an error/stock warning is shown.

Test Procedure	1. Add an in-stock product to your cart. 2. Check the cart items and totals. 3. Increase stock quantity and update totals. 4. Test quantity > stock and confirm rejection notice. 5. Remove the item and check the cart for adjustments.
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Table 4.5: TC-05 Cart Operations

4.3.6 **TC-6: Checkout Order Creation + Stock Recheck + Inventory Decrement**

Table 4.6 shows the test scenario for checkout and order placement. It validates delivery input fields, creates orders, clears carts, and adjusts inventories during order confirmation. It ensures the system blocks checkout when stock is insufficient and displays a suitable error message.

Test Case: TC-6	Checkout Order Creation + Stock Recheck + Inventory Decrement
Purpose	Verify checkout validation, order creation, cart clearing, and atomic stock decrement behavior.
Inputs	Delivery details (valid/invalid). Payment option (gateway or COD/simulated). Cart with in-stock items. Cart with insufficient stock scenario.
Pass/Fail Criteria	<u>Pass:</u> Valid checkouts make orders, reduce stock, and clear carts. Insufficient stock blocks were ordered. <u>Fail:</u> Orders created without stock updates, partial updates, or invalid deliveries accepted.
Expected Outputs	<u>Success:</u> order ID shown. Order stored. Inventory updated. Cart cleared. <u>Failure:</u> validation/stock error shown. No order confirmed.
Test Procedure	1. Add products to the cart. 2. Checkout and enter valid delivery information. 3. Confirm order, verify stock, and clear cart. 4. Repeat with incorrect fields and check for validation problems. 5. Repeat if stock is insufficient and ensure order is blocked.

Table 4.6: TC-06 Checkout Order Creation

4.3.7 **TC-7: Order History Access Control + Listing**

Table 4.7 shows a test case for order history and access control. This ensures that customers may only access their own orders and see accurate order status information. Additionally, it prevents unwanted access to other users' orders.

Test Case: TC-7	Order History Access Control + Listing
Purpose	Verify that customers can view only their own orders and correct statuses.
Inputs	Logged-in customer. Orders belonging to same customer. Orders belonging to a different customer (negative test).
Pass/Fail Criteria	<u>Pass</u> : only own orders displayed. No access to other users' orders. <u>Fail</u> : other users' orders visible or query returns unauthorized data.
Expected Outputs	<u>Success</u> : Order history displays only the logged-in customer's orders with correct current status. <u>Failure</u> : Unauthorized orders are not displayed. Access is denied or data is hidden if an invalid request is made.
Test Procedure	1. Log in as Customer A and open Order History. 2. Confirm that only Customer A orders display. 3. Check if access to Customer B's order is blocked or hidden.

Table 4.7: TC-07 Order History Access Control + Listing

4.3.8 TC-8: Profile Update Validation + Persistence

Table 4.8 shows the test scenario for profile updates. This ensures that editable profile fields are evaluated, saved correctly, and remain consistent upon reloading the screen. It also rejects invalid values and provides clear error feedback.

Test Case: TC-8	Profile Update Validation + Persistence
Purpose	Verify profile editing validation and that updates persist correctly.
Inputs	Update name/phone/address fields (valid). Invalid inputs (empty required fields, invalid phone).
Pass/Fail Criteria	<u>Pass</u> : valid updates saved and shown on reload. Invalid updates rejected with message. <u>Fail</u> : invalid data saved or valid data did not persist.
Expected Outputs	<u>Success</u> : Profile updates are saved and shown correctly after refresh/reopen. <u>Failure</u> : Invalid inputs are rejected with a message and previous valid data remains unchanged.
Test Procedure	1. Open the Profile screen. 2. Update valid fields and save. 3. Reopen Profile and confirm persistence. 4. Enter invalid values and confirm validation blocks save.

Table 4.8: TC-08 Profile Update Validation + Persistence

4.3.9 TC-9: Admin Product CRUD + Field Validation

Table 4.9 presents the test case for admin product management. It verifies product creation, updating, and deletion along with required field validation. It also checks that changes are reflected correctly in the customer catalog view.

Test Case: TC-9	Admin Product CRUD + Field Validation
Purpose	Verify admin can add/update/delete products and that required fields are enforced.
Inputs	Add product (valid fields). Add product (missing required fields). Update product (price/description/category). Delete product.
Pass/Fail Criteria	<u>Pass:</u> CRUD works for admin. Invalid product data rejected. <u>Fail:</u> invalid data accepted or CRUD fails without feedback.
Expected Outputs	<u>Success:</u> Admin can add/update/delete products and changes reflect in customer catalog. <u>Failure:</u> Invalid product data is rejected with an error and no incorrect product record is saved.
Test Procedure	1. Login as admin and open Product Management. 2. Create products with valid fields and verify them appear in catalog. 3. Try to create/update missing fields and verify validation errors. 4. Update existing products and verify changes. 5) Delete a product and verify removal.

Table 4.9: TC-09 Admin Product CRUD + Field Validation

4.3.10 TC-10: Admin Media Upload + Product Linking

Table 4.10 presents the test case for admin media management. It verifies uploading product images and 3D model assets and correctly linking them to a product record. It also ensures that images appear in product details and that AR becomes available only after a valid model is associated.

Test Case: TC-10	Admin Media Upload + Product Linking
Purpose	Verify that admin uploads media and correctly links it to a product for display and AR availability.
Inputs	Upload image (valid/invalid format). Upload 3D model (valid/invalid). Select target product.

Pass/Fail Criteria	<u>Pass:</u> media uploads succeed. Product upgrades include media references. The customer sees visuals. AR becomes available once a model exists. <u>Fail:</u> missing links, inaccessible media, and AR enabled without a proper model reference.
Expected Outputs	<u>Success:</u> Images and 3D models were successfully uploaded and connected to the product. Product information include images, and AR is available after linking the model. <u>Failure:</u> Uploads that are unsupported or fail are denied with an error message, and broken links are not preserved.
Test Procedure	1. Log in as administrator and select a product. 2. Upload product photos and verify them. displayed on the client product detail screen. 3. Upload/associate 3D model and confirm AR indication availability. 4. Test invalid uploads and confirm rejection messages.

Table 4.10: TC-10 Admin Media Upload + Product Linking

4.3.11 TC-11: Admin Stock Update Rules (Non-negative + Reflected)

Table 4.11 shows a test case for inventory/stock management. It ensures that stock changes are only allowed for non-negative values, and that setting stock to zero indicates out-of-stock status. It also ensures that customer carts and checkout flows reflect current supply levels.

Test Case: TC-11	Admin Stock Update Rules
Purpose	Verify stock updates are validated and reflected in customer views/cart constraints.
Inputs	Set stockQuantity to valid integer. Set stockQuantity to negative/non-numeric. Set stock to 0 (out-of-stock).
Pass/Fail Criteria	<u>Pass:</u> stock updates saved. Negative values blocked. Out-of-stock reflected correctly. <u>Fail:</u> negative stock saved or customer can exceed stock in cart.
Expected Outputs	<u>Success:</u> Stock updates are saved appropriately. When stock is set to 0, it indicates that the item is out of stock, and the cart/checkout will change accordingly. <u>Failure:</u> Stock values that are negative or incorrect are rejected. Stock state does not become inconsistent.
Test Procedure	1. Log in as an admin and update product stock to a valid value. 2. Ensure that clients see the accurate stock status. 3. Set stock to 0 and test out-of-stock behavior.

	4. Test negative stock and confirm that the update is denied.
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Table 4.11: TC-11 Admin Stock Update Rules

4.3.12 **TC-12: Admin Order Status Update + Customer Visibility**

Table 4.12 shows the test scenario for administrative order processing. This ensures that the administrator can update order status during its lifecycle and maintain the most recent state. This validates that clients can view updated order statuses in their history.

Test Case: TC-12	Admin Order Status Update + Customer Visibility
Purpose	Verify admin can update order status and customer sees the updated status in order history.
Inputs	Select existing order. Change status (pending to confirmed to shipped to delivered). Optional cancellation/refund states if supported in the build.
Pass/Fail Criteria	<u>Pass</u> : status update persists. Customer sees updated status. <u>Fail</u> : status does not persist, inconsistent states, or customer view not updated.
Expected Outputs	<u>Success</u> : The admin status is successfully updated and shown to the customer in their order history. <u>Failure</u> : Invalid or unsuccessful status updates result in an error message. The customer view remains consistent.
Test Procedure	1. Place your order as a customer. 2. Log in as an administrator and open Order Management. 3. Update the status step by step and confirm the saving. 4. Log in as a customer and check the new status in Order History.

Table 4.12: TC-12 Admin Order Status Update

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